ARTICLE OPEN

Check for updates

Shared and disorder-specific computational mechanisms of interference and response inhibition in schizophrenia and major depressive disorder

Tongyi Zhang[1], Sirui Wang[1], Haoyang Shi[1], Jia Li[1], Xiaolong Yang[2], Jin Jiao[3], Xiaoning Huo[2] and Xin Zhao[1]✉

© The Author(s) 2026

Inhibitory control deficits are consistently observed in both schizophrenia (SCZ) and major depressive disorder (MDD), yet whether these impairments reflect shared tansdiagnostic mechanisms or disorder-specific dysfunction remains poorly understood. We recruited $N = 259$ participants ($n = 87$ SCZ, $n = 86$ MDD, $n = 86$ healthy controls) who completed *Stroop* and *Go/No-Go* tasks assessing interference and response inhibition, respectively. Hierarchical drift-diffusion modeling (HDDM) was used to decompose behavioral performance into latent computational parameters. Both patient groups exhibited significant behavioral impairments across most measures, with SCZ showing greater severity. At the computational level, reduced drift rate—reflecting less efficient evidence accumulation—emerged as a core transdiagnostic deficit across both inhibitory subcomponents and both disorders, with more pronounced reductions in SCZ. By contrast, prolonged non-decision time was disorder-differentiating, being consistently elevated across both tasks in SCZ but, in MDD, confined to response inhibition and not robust to processing-speed adjustment. Within-group clinical correlates further diverged, relating to negative-symptom severity and illness duration in SCZ and to current depressive symptom severity in MDD, although these diagnosis-specific associations were not directly compared across disorders. Machine learning classification corroborated these findings, with drift rate serving as the primary shared contributor to patient–control discrimination (area under the receiver operating characteristic curve [AUC] = 0.762–0.846), and non-decision time on the interference task the top-ranked feature in the SCZ–MDD classifier (AUC = 0.718). Together, these findings delineate shared versus disorder-specific computational signatures of inhibitory control impairment in SCZ and MDD, and suggest that computational phenotyping of inhibitory control may serve as a quantitative adjunct in precision psychiatry.
Preregistration: This study was preregistered with the Chinese Clinical Trial Registry prior to data collection (registration number: ChiCTR2500111117; https://www.chictr.org.cn/showproj.html?proj=290089).

Translational Psychiatry (2026)16:590 ; https://doi.org/10.1038/s41398-026-04345-3

INTRODUCTION

Cognitive impairment is increasingly recognized as a core feature of both schizophrenia (SCZ) and major depressive disorder (MDD) [1, 2]. However, whether the cognitive deficits observed in these two disorders arise from shared or distinct underlying mechanisms remains unclear. To date, research has largely focused on describing cognitive impairment phenotypes—such as global executive dysfunction or attentional deficits—while fine-grained cognitive markers that can reliably distinguish diagnoses or predict treatment response remain insufficiently established [3], although preliminary work in MDD suggests that inhibitory-control-related neural activation may predict treatment response [4]. This has hindered the identification of precise intervention targets and the development of individualized treatment strategies. Addressing this gap calls for two complementary advances: a theoretical framework that moves beyond categorical diagnoses to examine the dimensional structure of cognitive processes, and analytic tools capable of resolving aggregate behavioral indices into the latent decision subprocesses that generate them.

In this regard, the Research Domain Criteria (RDoC) framework offers a valuable theoretical perspective, encouraging a shift in psychopathology research from diagnosis-centered approaches toward transdiagnostic, dimensional investigations of underlying mechanisms [5, 6]. Within this framework, SCZ and MDD may share certain pathophysiological mechanisms across several cognitive dimensions. Indeed, the two disorders converge on shared neurobiological substrates at multiple levels, including neurotransmitter systems (e.g., dopamine), brain structural changes (e.g., reduced hippocampal volume), and genetic susceptibility [7–9]. Importantly, these transdiagnostic commonalities do not preclude disorder-specific impairments; rather, they highlight the need to identify cognitive processes that reflect general vulnerability as well as those that distinguish the two disorders. Clarifying this distinction remains an important prerequisite for precision diagnosis and stratified intervention [10].

Within the RDoC framework, inhibitory control can be examined as a transdiagnostic cognitive process across conventional diagnostic categories [11]. Inhibitory control refers to the ability to actively suppress irrelevant stimuli, interfering mental processes, and

[1]School of Psychology, Northwest Normal University, Lanzhou, China. [2]The Third People's Hospital of Lanzhou, Lanzhou, China. [3]The Third People's Hospital of Tianshui, Tianshui, China. ✉email: psyzhaoxin@nwnu.edu.cn

Received: 7 February 2026 Revised: 15 July 2026 Accepted: 24 July 2026
Published online: 31 July 2026

prepotent behavioral responses [12]. It is subserved by prefrontal–cingulate circuits and related prefrontal–striatal–thalamic networks [13, 14]. At the paradigm level, inhibitory control deficits in SCZ have been consistently observed across *stop-signal* and *Stroop* tasks [15, 16], and have also been reported using *Go/No-Go* paradigms [17]. Conceptually, inhibitory control is not a unitary construct. The literature commonly distinguishes response inhibition from interference control while recognizing their overlap; in the present study, the latter is operationalized as interference inhibition [12, 18]. Response inhibition is typically assessed using paradigms such as the *Go/No-Go* and *stop-signal* tasks, whereas interference inhibition is commonly measured by conflict tasks such as the *Stroop* and *flanker* tasks [13, 19]. Because these subcomponents may be differentially affected across disorders, examining both within the same study can reveal whether SCZ and MDD diverge in their patterns of response- versus interference-inhibition deficits.

Because SCZ and MDD have distinct pathophysiological profiles, they may also show different patterns of computational impairment on inhibitory control tasks. In SCZ, abnormal dopamine signaling—particularly elevated presynaptic striatal dopamine function—is a well-established pathophysiological feature [20]. Behavioral and neuroimaging studies have reported prolonged *stop-signal* reaction times and reduced proactive inhibition in SCZ [21, 22]. In contrast, cognitive impairment in MDD is closely linked to difficulties in emotion regulation and negative cognitive biases. Patients with MDD tend to over-process negative, self-relevant, or failure-related information and are slow to redirect cognitive resources toward current task goals [23]. This pattern of prolonged engagement with negative material and slowed disengagement suggests a candidate account that the present study examines empirically rather than treats as a fixed prediction: patients with MDD might adopt more conservative decision strategies that require more evidence before committing to a response, which could manifest computationally as a higher response threshold or as systematic biases in evidence integration, rather than as a global reduction in processing efficiency. Consistent with this account, prior studies have reported altered allocation between anticipatory (proactive) and stimulus-driven (reactive) control processes, as well as abnormal conflict adaptation, in MDD [24, 25].

However, adjudicating among competing disorder-specific computational hypotheses goes beyond what traditional behavioral measures alone can resolve. Reaction time and accuracy are composite outcomes reflecting the combined influence of multiple cognitive subprocesses, including stimulus encoding, evidence accumulation, decision-threshold setting, and response execution [26]. When two patient groups show prolonged reaction times and elevated error rates, the underlying sources of their impairments may differ in ways that aggregate behavioral indices cannot reveal, obscuring the true locus of deficit. The hierarchical drift-diffusion model (HDDM) directly addresses this decomposition problem. Grounded in sequential sampling theory, HDDM decomposes reaction time distributions and response accuracy into latent parameters with clear psychological interpretations [26, 27]. The core parameters include drift rate (v), reflecting the efficiency of evidence accumulation; decision boundary (a), reflecting response caution and the amount of evidence required before a response; non-decision time (T_{er}), reflecting the duration of perceptual encoding and motor execution; and starting point (z), reflecting a prior response tendency [26, 28].

Existing neuroimaging evidence indicates that the neural substrates of inhibitory control are partly task-dependent across *Go/No-Go* and *stop-signal* paradigms [13, 29, 30]; however, these studies do not establish a specific HDDM-parameter pattern for SCZ. Several diffusion-model studies have nonetheless reported reduced drift rates in schizophrenia and psychosis across working-memory and cognitive-control tasks [31, 32], consistent with meta-analytic evidence that processing-speed deficits are among the most robust cognitive impairments in this disorder [33]. By contrast, diffusion-model studies in MDD and depressive symptoms have reported task-dependent abnormalities, including reduced drift rates and altered emotional-information processing [34, 35]; accordingly, the present study does not prespecify a single MDD parameter signature. HDDM and related diffusion models have been applied to several psychiatric conditions, including major depressive disorder [35], social anxiety disorder [36], schizophrenia [37, 38], and attention-deficit/hyperactivity disorder [39], with converging evidence reviewed elsewhere [40].

In the present study, we employed HDDM to model the performance of patients with SCZ, patients with MDD, and healthy controls (HCs) on interference- and response-inhibition tasks, aiming to characterize the latent computational mechanisms underlying inhibitory control impairments in both disorders and to identify their transdiagnostic and disorder-specific features (Fig. 1). The analysis proceeded from behavioral characterization and computational decomposition through clinical relevance to individual-level discriminability. Four aims guided the analyses. Aim 1 was to characterize which HDDM parameters are abnormal in each disorder and whether these abnormalities are transdiagnostic or disorder-specific; we hypothesized that drift-rate reductions relative to HCs would be more pronounced in SCZ than in MDD, whereas for MDD we did not commit to a directional prediction, treating decision boundary, starting point, and drift rate as competing candidate signatures. Aim 2 was to test whether interference and response inhibition draw on shared or separable computational resources, and whether this structure is altered in the patient groups, using cross-task parameter consistency and task-type classification. Aim 3 was to examine whether parameter abnormalities relate to symptom severity within each diagnostic group, with any cross-disorder divergence treated as hypothesis-generating given the non-identical symptom instruments. Aim 4 was to test whether HDDM parameters carry individual-level diagnostic information across the three groups.

METHODS

Ethics statement

The Ethics Committee of the Third People's Hospital of Lanzhou (approval number: TPHL2025014) and the Ethics Committee of Northwest Normal University (approval number: NWNU20251009) approved this study, which was conducted according to the Declaration of Helsinki. All participants provided written informed consent before participation.

Participants

We enrolled 259 participants during the study execution period (October–December 2025), comprising 87 patients with SCZ, 86 patients with MDD, and 86 HCs, recruited at the Third People's Hospital of Lanzhou. The two patient groups were enrolled consecutively from the available clinical pool without a sex-balancing scheme, whereas the HCs were recruited to be comparable to the patients in age, sex, and years of education. Although sex did not moderate schizotypy–executive-control associations in one schizophrenia-spectrum study [41], sex differences are well documented in the clinical presentation of MDD and SCZ [42, 43]; sex was therefore included as a covariate in the behavioral analyses of covariance (ANCOVAs) of inhibitory control measures and in the partial correlations between HDDM parameters and behavioral and clinical indices. Sample size was primarily determined by the requirements of hierarchical Bayesian parameter estimation, which are larger than those of conventional behavioral analyses [27]; accordingly, we aimed for 85–90 participants per group, also allowing for potential attrition. For reference, an a priori calculation using G*Power 3 [44] indicated that approximately 53 participants per group would suffice to detect a medium behavioral effect ($f = 0.25$, $\alpha = 0.05$, $1 - \beta = 0.80$). The study was preregistered with the Chinese Clinical Trial Registry before data collection (registration number: ChiCTR2500111117; https://www.chictr.org.cn/showproj.html?proj=290089).

Patients with SCZ and MDD received a confirmed diagnosis based on the Structured Clinical Interview for DSM-IV Axis I Disorders (SCID-I) [45], administered by trained psychiatrists. The diagnoses were subsequently

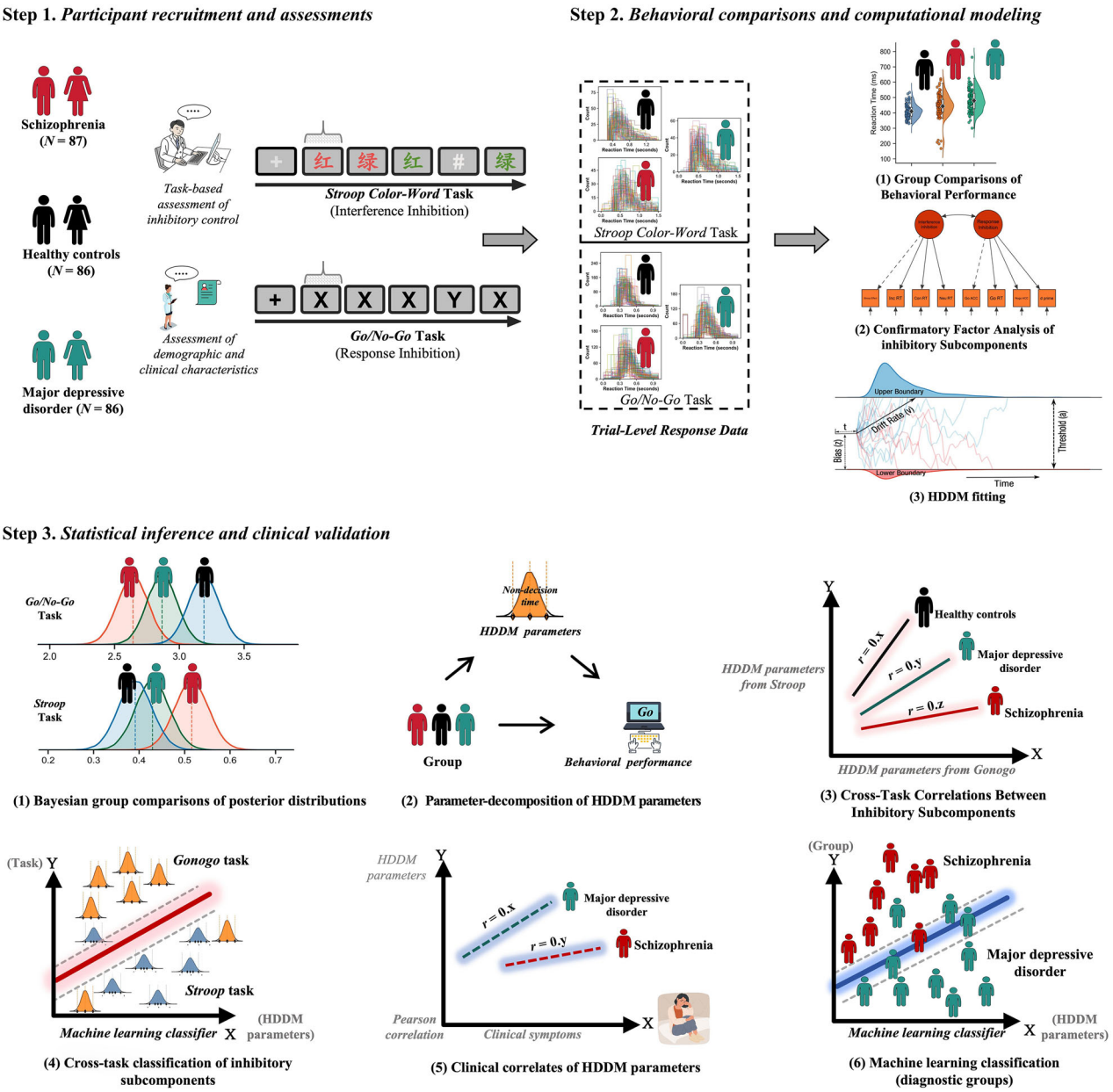


Fig. 1 Study design and analytical framework. Step 1 (participant recruitment and assessments) shows enrollment of 87 patients with schizophrenia (SCZ), 86 with major depressive disorder (MDD), and 86 healthy control (HC) participants matched for age, sex, and years of education, together with the *Stroop* and *Go/No-Go* task structures used to elicit interference inhibition and response inhibition. Step 2 (behavioral comparisons and computational modeling) integrates (i) group comparisons of behavioral performance, (ii) confirmatory factor analysis (CFA) of inhibitory subcomponents, and (iii) hierarchical drift-diffusion model (HDDM) fitting on trial-level response data. Step 3 (statistical inference and clinical validation) comprises six analyses, mapped to the four pre-specified Aims: (1) Bayesian group comparisons on posterior parameter distributions, characterizing transdiagnostic and disorder-specific HDDM parameter differences (Aims 1–2); (2) parameter-decomposition analysis quantifying how HDDM parameters account for within-group condition-to-performance effects (Aims 1–2); (3) within-subject cross-task correlations between *Stroop* and *Go/No-Go* HDDM parameters, indexing whether the two inhibitory subcomponents share computational resources (Aims 1–2); (4) subject-level cross-task classification testing whether the two inhibitory subprocesses remain computationally separable on the basis of HDDM parameters (Aims 1–2); (5) partial-correlation analysis of HDDM parameters with clinical symptom measures within each patient group, controlling for age, sex, education, and medication dose (Aim 3); and (6) support vector machine (SVM) classification of diagnostic groups based on HDDM parameters, with SHAP feature importance and demographic-control sensitivity analyses (Aim 4). CFA confirmatory factor analysis, HC healthy control, HDDM hierarchical drift-diffusion model, MDD major depressive disorder, SCZ schizophrenia, SHAP SHapley additive exPlanations, SVM support vector machine.

cross-mapped to ICD-10 codes F20.0–F20.9 (SCZ) and F32–F33 (MDD) for compatibility with the ICD-10-based hospital electronic medical record system used in China. For the present study, clinical stability included a stable medication regimen for ≥ 6 weeks in SCZ, consistent with prior work [46], and ≥ 2 weeks in MDD as a study-specific criterion. Patients with SCZ were excluded if they met current or lifetime SCID-I criteria for any other Axis I disorder (mood, anxiety, eating, somatoform, or adjustment disorders), substance use disorder, or intellectual disability. Patients with MDD were excluded if they met current or lifetime SCID-I criteria for any schizophrenia-spectrum disorder, bipolar disorder, or other Axis I disorder, including MDD with psychotic features (ICD-10 F32.3/F33.3), substance use disorder, or intellectual disability. No participant with MDD in the final

Table 1. Participants' demographics and clinical characteristics.

Variable	HC ($N = 86$)	MDD ($N = 86$)	SCZ ($N = 87$)	Statistic	p-value
Demographic characteristics					
Age (years)	35.86 (11.78)	38.34 (12.56)	36.72 (11.48)	$F = 1.14$	0.322
Sex	M = 43, F = 43	M = 41, F = 45	M = 47, F = 40	$\chi^2 = 1.18$	0.554
Ethnicity	Han = 79, Other = 7	Han = 78, Other = 8	Han = 78, Other = 9	$\chi^2 = 0.52$	0.774
Education (years)	12.28 (3.54)	11.82 (3.68)	11.18 (3.52)	$F = 2.06$	0.130
RPM	48.52 (8.68)	43.86 (9.42)	42.42 (10.76)	$F = 11.86$	< **0**.001 [a]
SES	24.12 (6.18)	23.86 (6.72)	23.51 (6.35)	$F = 0.24$	0.786
BMI	23.24 (3.58)	23.38 (3.82)	24.82 (4.18)	$F = 4.12$	**0**.017 [b]
Only child, *n* (%)	14 (16.3%)	15 (17.4%)	12 (13.8%)	$\chi^2 = 0.48$	0.786
Clinical characteristics					
Age at onset (years)	–	30.86 (10.24)	27.00 (9.86)	$t(171) = 2.52$	**0**.013
Duration of illness (years)	–	7.48 (4.16)	9.72 (6.38)	$t(171) = -2.73$	**0**.007
Family history, *n* (%)	–	19 (22.1%)	24 (27.6%)	$\chi^2 = 0.71$	0.398
Antidepressant, *n* (%)	–	78 (90.7%)	23 (26.4%)	$\chi^2 = 72.46$	< **0**.001
Antipsychotic, *n* (%)	–	18 (20.9%)	87 (100%)	$\chi^2 = 113.82$	< **0**.001
CPZ equivalent (mg/day)	–	152.36 (98.42)	418.36 (228.74)	$t(103) = -7.49$	< **0**.001 [c]
FLU equivalent (mg/day)	–	35.84 (14.62)	24.28 (12.36)	$t(99) = 3.62$	< **0**.001 [c]
RSESE positive, *n* (%)	–	1 (1.2%)	6 (6.9%)	–	0.117 [d]
HAMD-17					
HAMD-17 total	–	23.02 (4.58)	13.30 (5.18)	$t(171) = 13.07$	< **0**.001
HAMD anxiety/somatization	–	7.86 (2.14)	4.28 (1.92)	$t(171) = 11.58$	< **0**.001
HAMD weight	–	1.24 (0.68)	0.86 (0.72)	$t(171) = 3.57$	< **0**.001
HAMD cognitive disturbance	–	4.52 (1.86)	2.64 (1.48)	$t(171) = 7.36$	< **0**.001
HAMD diurnal variation	–	1.36 (0.82)	0.92 (0.76)	$t(171) = 3.65$	< **0**.001
HAMD retardation	–	4.18 (1.64)	2.42 (1.36)	$t(171) = 7.68$	< **0**.001
HAMD sleep disturbance	–	3.86 (1.42)	2.18 (1.28)	$t(171) = 8.16$	< **0**.001
HAMA-14					
HAMA total	–	18.12 (5.36)	10.28 (4.72)	$t(171) = 10.21$	< **0**.001
HAMA psychic anxiety	–	10.24 (3.18)	5.86 (2.64)	$t(171) = 9.85$	< **0**.001
HAMA somatic anxiety [e]	–	7.88 (2.86)	4.42 (2.38)	$t(171) = 8.65$	< **0**.001
PANSS					
PANSS positive	–	–	15.42 (5.36)	–	–
PANSS negative	–	–	19.84 (6.28)	–	–
PANSS general	–	–	35.16 (8.42)	–	–

Data are *M* (*SD*) or *n* (%). Demographic variables were compared across the three groups by one-way ANOVA (*F*) or chi-squared test (χ^2); clinical variables were compared between MDD and SCZ by independent-samples *t*-test or chi-squared test. Fisher's exact test was used where expected cell frequencies were < 5, in which case no test statistic is reported. *p*-values are uncorrected and descriptive; FDR correction was applied within each pre-specified hypothesis family in subsequent analyses (see Methods). Bold indicates $p < 0.05$. SES was computed as a composite index (Supplementary Methods).

ANOVA analysis of variance, *BMI* body mass index, *CPZ* chlorpromazine, *F* female, *FDR* false discovery rate, *FLU* fluoxetine, *HAMA-14* 14-item Hamilton anxiety rating scale, *HAMD-17* 17-item Hamilton depression rating scale, *HC* healthy control, *M* male, *MDD* major depressive disorder, *PANSS* positive and negative syndrome scale, *RPM* Raven's progressive matrices, *RSESE* rating scale for extrapyramidal side effects, *SCZ* schizophrenia, *SD* standard deviation, *SES* socioeconomic status.

[a]Post hoc: HC > SCZ ($p < 0.001$), HC > MDD ($p = 0.003$); MDD vs. SCZ ($p = 0.346$).

[b] Post hoc: SCZ > HC ($p = 0.016$); SCZ vs. MDD ($p = 0.055$) and MDD vs. HC ($p = 0.813$) were not significant.

[c]Medication-equivalent doses were compared only among patients receiving the respective drug class (antipsychotics: MDD $n = 18$, SCZ $n = 87$, total $N = 105$, $df = 103$; antidepressants: MDD $n = 78$, SCZ $n = 23$, total $N = 101$, $df = 99$).

[d]RSESE positivity: any score ≥ 1 versus 0. Compared by Fisher's exact test. The RSESE was not administered to HCs, who were not on psychotropic medication.

[e]HAMA somatic anxiety scores were derived as HAMA total − HAMA psychic anxiety.

sample met SCID-I criteria for current or lifetime psychotic features. HCs were recruited through community outreach and online advertisements and screened with the Mini-International Neuropsychiatric Interview (MINI) [47] to exclude individuals with current or past psychiatric disorders. Supplementary Table S1 provides detailed inclusion and exclusion criteria for all groups.

Three supplementary indices were used to characterize the sample. Socioeconomic status (SES) was assessed using a comprehensive questionnaire covering material wealth, parental education, parental occupation, and household income (see Supplementary Methods). Given the well-documented associations between SES and cognitive performance [48, 49], SES was treated as a matching variable rather than a statistical control, and the three groups did not differ on this index (Table 1). Raven's Progressive Matrices (RPM) indexed nonverbal fluid intelligence, and body mass index (BMI) captured potential metabolic effects of antipsychotic treatment. Neither was retained as a primary covariate (see Supplementary Methods). For RPM, sensitivity analyses with RPM as a conservative covariate are reported in Supplementary Table S2.

Clinical symptom assessment

Both patient groups completed assessments of depressive and anxiety symptoms to characterize affective symptom severity [50]. The 17-item Hamilton Depression Rating Scale (HAMD-17) was used to assess depressive symptom severity [51], with six factor scores derived from the classical six-factor structure. Anxiety symptom severity was evaluated using the 14-item Hamilton Anxiety Rating Scale (HAMA-14) [52], with two factor scores computed following established guidelines. For the SCZ group, the Positive and Negative Syndrome Scale (PANSS) [53], comprising positive, negative, and general psychopathology subscales, was used to assess psychotic symptom severity. All clinical assessments were performed by trained psychiatrists. Additional clinical information, including illness duration, number of hospitalizations, current medication regimen, and treatment history, was obtained from electronic medical records. The doses of antipsychotics and antidepressants were converted to chlorpromazine [54] and fluoxetine equivalents [55], respectively, to facilitate cross-study comparisons and control for potential medication effects (detailed medication profiles in Supplementary Tables S3 and S4). Extrapyramidal symptoms (EPS) in the two patient groups were evaluated at the time of clinical assessment using the Rating Scale for Extrapyramidal Side Effects (RSESE; also known as the Simpson–Angus Scale) [56].

Inhibitory control behavioral tasks

We employed the *Stroop* and *Go/No-Go* tasks—two paradigms that tap distinct subcomponents of inhibitory control, with well-characterized neural substrates and a long history of application in psychiatric research [3, 12, 14, 29, 57]—to assess interference inhibition and response inhibition, respectively. A priori, we selected standard color-word *Stroop* and *Go/No-Go* variants with emotionally neutral stimuli to isolate cold cognitive control from emotion-driven interference, consistent with established distinctions between hot and cold cognition [58, 59].

Stroop task. Participants responded to the font color while ignoring word meaning (Fig. 1) [60]. The task included congruent, incongruent, and neutral conditions, with the *Stroop* interference effect (the difference in reaction time between incongruent and congruent conditions) as the primary outcome. The task comprised one practice block of 18 trials, followed by 108 test trials distributed across three blocks.

Go/No-Go task. Participants responded to Go stimuli while withholding responses to No-Go stimuli (Go:No-Go ratio = 75:25; Fig. 1) [29]. Primary outcomes included Go and No-Go accuracy, Go reaction time, and the discriminability index d' [61]. The task comprised one practice block of 20 trials (15 Go, 5 No-Go, preserving the test-phase ratio), followed by 400 test trials distributed across four blocks.

For both tasks, participants were required to reach ≥ 85% accuracy during practice before proceeding to the test phase. This criterion was designed to establish a common minimum competence threshold across the groups and limit early task-acquisition effects on HDDM parameter estimation [62]. We empirically examined the influence of practice on subsequent HDDM parameter estimation through partial correlations between practice-phase performance and HDDM parameters controlling for diagnostic group (see Supplementary Results), which indicated that group-level HDDM differences were unlikely to be primarily attributable to differential practice-related acquisition. Each task lasted approximately 15 min. Detailed task parameters are provided in Supplementary Methods, split-half reliability estimates are reported in Supplementary Table S6, and task schematics are shown in Fig. 1.

Procedure

All participants completed the full assessment battery within one week of enrollment. The first part involved clinical assessment (patient groups only), which included HAMD-17 and HAMA-14 (and the PANSS for the SCZ group) administration and demographic and clinical information collection from electronic medical records (approximately 40 min). The second part involved behavioral task testing: participants completed the *Stroop* task first, followed by a brief rest and the *Go/No-Go* task (task order was held constant across participants to minimize practice-transfer variability). Behavioral task assessors were blinded to participants' diagnostic-group status, and participants were not informed of the specific study hypotheses, whereas data analysts had access to diagnostic-group labels. Total behavioral testing time was approximately 30 min. All behavioral tasks were programmed using E-Prime 3.0 (Psychology Software Tools, Pittsburgh, PA, USA) and presented on a 17.3-inch monitor (resolution: 1920 × 1080; refresh rate: 60 Hz).

Data analysis

Behavioral and clinical analyses. One-way ANCOVAs were conducted to examine group differences (SCZ, MDD, and HC) in inhibitory control measures, with age, sex, and years of education used as covariates. Significant main effects were further examined using Tukey's honestly significant difference (HSD) post hoc comparisons. Within each patient group, partial correlation analyses examined the associations between inhibitory control performance and clinical variables (e.g., illness duration, PANSS subscale scores, HAMD-17 scores, and HAMA-14 scores), controlling for age, sex, years of education, and medication dose (chlorpromazine equivalents for SCZ; fluoxetine equivalents for MDD). The same covariate set, including medication dose, was applied to all symptom-related correlations involving HDDM parameters. Distributional assumptions were evaluated for ANCOVAs, partial correlations, and *t* tests using the relevant variables or model residuals; Levene's test was used where applicable, and alternative tests were applied when assumptions were violated. All frequentist tests were two-sided, and all multiple comparisons were corrected using the Benjamini–Hochberg false discovery rate (FDR) procedure [63].

Confirmatory factor analysis. We employed confirmatory factor analysis (CFA) to establish the measurement-structure validity of the two inhibitory subcomponents and to test whether their factor structure differed across the three groups [64]. Observed indicators were drawn from both tasks: four from the *Stroop* task (reaction times for congruent, incongruent, and neutral conditions, and the interference effect) and four from the *Go/No-Go* task (Go accuracy, No-Go accuracy, Go reaction time, and d'). A single-factor model (all indicators loading onto a unitary "inhibitory control" factor) was compared with a two-factor model (*Stroop* indicators loading onto an "interference inhibition" factor and *Go/No-Go* indicators loading onto a "response inhibition" factor). The comparative fit index (CFI), χ^2, the Tucker–Lewis index (TLI), the root mean square error of approximation (RMSEA) with 90% confidence intervals (CIs), and the standardized root mean square residual (SRMR) were used to evaluate model fit [65]. Model comparisons used chi-square difference tests and the Akaike information criterion (AIC). Then, the measurement invariance of the inhibitory control factor structure across the three groups was tested using multi-group CFA [66], sequentially examining configural, metric, and partial scalar invariance. Invariance was evaluated using nested-model chi-square difference tests and CFI changes ($|\Delta CFI| \leq 0.010$ indicated invariance) [65]. All CFAs were conducted using the lavaan package (v0.6–17) in R (v4.3.1) [67].

HDDM analysis

Data preprocessing: Trial-by-trial behavioral data from the *Stroop* and *Go/No-Go* tasks were analyzed using the HDDM package (v1.0.1) [27]. For the *Stroop* task, consistent with standard practice in HDDM applications [26], only correct-response trials were included. For the *Go/No-Go* task, a stimulus-coding approach was adopted for drift-diffusion modeling [27], treating the task as a binary stimulus classification decision process (see Supplementary Methods). In this parameterization, the upper boundary represented "Go" classification decisions, and the lower boundary represented "No-Go" classification decisions. Given the base-rate asymmetry of the Go stimuli (75%), the model included a starting point bias parameter (z) to capture participants' prior response tendency toward the Go stimuli [68]. Data coding followed HDDM documentation recommendations (see Supplementary Methods). The model specified 5% of trials as potential outliers ($p_{outlier} = 0.05$) to account for contaminant responses [28].

Model specification: Three hierarchical model classes were specified (Supplementary Table S7): (1) a baseline model (M0) with all four core drift-diffusion model (DDM) parameters (a, v, T_{er}, z) and inter-trial variability parameters (s_v, s_z, s_t) estimated jointly as the full baseline model; (2) a hierarchical model (M1) allowing a, v, T_{er}, and z to vary across the groups; and (3) a mixed-effects regression family comprising four submodels (M2a, M2v, M2T_{er}, M2z), each allowing the corresponding parameter to be modeled as a function of group, adopted as our primary model class because it estimated group differences while accounting for individual variability [27, 69]. Default prior distributions from the HDDM package were employed [28]. Model comparisons based on the deviance information criterion (DIC) are reported in Supplementary Table S10.

Model fitting and diagnostics: Model parameters were estimated within a hierarchical Bayesian framework, with posterior distributions obtained via Markov chain Monte Carlo (MCMC) sampling. Each model was run with four independent MCMC chains, each comprising 15,000 samples, with the first 5000 discarded as burn-in [36]. Convergence was assessed using the Gelman–Rubin statistic ($\hat{R} < 1.01$) [70]. Model comparison was performed using DIC. Model adequacy was further evaluated via posterior predictive checks based on 500 simulated datasets generated using the *postpred_gen* function in HDDM [27].

Statistical inference and clinical association analyses. Group differences were examined using Bayesian inference based on posterior parameter distributions, with evidence strength quantified by Bayes factors (BF_{10}) [71]. Uncertainty in group differences was characterized using 95% highest density intervals (HDIs), with statistical decisions based on the relation between the 95% HDIs and the region of practical equivalence (ROPE) [72]. Beyond group comparisons, partial correlation analyses examined the associations between HDDM parameters and behavioral indices, controlling for age, sex, and years of education. For associations between HDDM parameters and clinical symptoms, medication dose was additionally included as a covariate, consistent with the scheme described above [69]. The same FDR correction was applied. As a robustness check, we re-ran the primary group comparisons including *Stroop* congruent-trial mean reaction time (RT) as a within-paradigm processing-speed proxy (see Supplementary Methods).

Machine learning classification. We constructed support vector machine (SVM)–based classification models to evaluate the diagnostic classification utility of HDDM parameters [46], implemented via scikit-learn (v1.3.0) in Python [73]. HDDM parameters from the *Stroop* and *Go/No-Go* tasks were used as input features for pairwise binary classification models (SCZ vs. HCs, MDD vs. HCs, and SCZ vs. MDD). Data were randomly split into training (80%) and test (20%) sets and evaluated on the held-out test set [74]. Classification performance was assessed using the area under the receiver operating characteristic curve (AUC), balanced accuracy (BAC), sensitivity, and specificity. The SHapley Additive exPlanations (SHAP) method [75] was used for feature importance analysis to identify the HDDM parameters that contributed most to the classification. Two SVM sensitivity analyses complemented the primary pairwise classifications [46]: first, a one-vs-rest multiclass SVM was trained for direct comparison with the pairwise approach; second, HDDM parameters were residualized separately for age, sex, years of education, and medication dose within each diagnostic group before SVM training. SHAP feature importance was computed for both the multiclass and pairwise models, with top-ranked features compared descriptively across HDDM parameters. Additional implementation details are provided in Supplementary Methods.

Examination of shared and disorder-specific mechanisms. We adopted the following complementary analytic strategies to further dissect shared and disorder-specific computational mechanisms (Fig. 1). First, parameter-decomposition analysis quantified the contribution of HDDM parameters to within-group condition-to-performance effects. Within each group, bootstrap procedures resampling participants with replacement (3000 resamples) estimated the decomposed contribution from task condition to behavioral performance via HDDM parameters. Because HDDM parameters are derived from the same RT and accuracy data, this analysis quantifies how the HDDM decomposes condition-to-performance variance rather than constituting a causal mediation test. Second, cross-task parameter consistency analysis tested whether the two inhibitory subcomponents share common computational resources. For each participant, corresponding HDDM parameters (v, a, T_{er}) were extracted from both tasks, followed by computing within-group Pearson correlations between *Stroop* and *Go/No-Go* estimates. Fisher's z-transformation was used to test between-group differences, with Benjamini–Hochberg FDR correction across the three pairwise contrasts for each parameter. Third, task-type classification analysis tested whether the two inhibitory subprocesses remain computationally separable. SVMs classified task type (*Stroop* versus *Go/No-Go*) based on subject-level HDDM parameters, with each participant contributing one parameter set per task, separately within each group. Because each participant contributed two non-independent observations, train–test splits were performed at the participant level. Feature ablation—sequential removal of individual parameters, in which ablated models are compared against the full model using bootstrap difference tests—quantified each parameter's contribution to task discrimination. This task-type SVM classified tasks based on within-subject parameter pairs, whereas the diagnostic SVM addressing Aim 4 classified diagnostic groups based on between-subject parameter profiles—two methodologically distinct analyses.

RESULTS

Demographic and clinical characteristics

The three groups did not differ significantly in age, sex distribution, years of education, socioeconomic status, ethnicity, or the proportion of only children (all $p > 0.05$; Table 1). Raven's Progressive Matrices scores differed significantly across the groups ($F(2, 256) = 11.86$, $p < 0.001$), with the HC group scoring higher than the SCZ ($p < 0.001$) and MDD ($p = 0.003$) groups; the difference between the two patient groups did not reach significance ($p = 0.346$). Body mass index also demonstrated a significant group difference ($F(2, 256) = 4.12$, $p = 0.017$), with the SCZ group showing higher BMI than the HCs ($p = 0.016$).

Regarding clinical characteristics, the SCZ group had a significantly earlier age at onset ($t(171) = 2.52$, $p = 0.013$) and longer illness duration ($t(171) = -2.73$, $p = 0.007$) than the MDD group. The MDD group had significantly higher HAMD-17 ($t(171) = 13.07$, $p < 0.001$) and HAMA-14 ($t(171) = 10.21$, $p < 0.001$) total scores than the SCZ group, with parallel patterns across all factor scores. The MDD group also showed a higher rate of antidepressant use than the SCZ group ($\chi^2(1) = 72.46$, $p < 0.001$) and higher fluoxetine-equivalent doses ($t(99) = 3.62$, $p < 0.001$), whereas the SCZ group showed a higher rate of antipsychotic use ($\chi^2(1) = 113.82$, $p < 0.001$) and higher chlorpromazine-equivalent doses ($t(103) = -7.49$, $p < 0.001$).

Detailed medication information is provided in Supplementary Tables S3 and S4. The SCZ sample was predominantly receiving atypical antipsychotic monotherapy, whereas the MDD sample was predominantly receiving first-line antidepressant monotherapy. Extrapyramidal symptoms, assessed by the RSESE as a binary indicator (positive: any score ≥ 1; absent: 0), were present in 7 of 173 patients (SCZ: 6/87; MDD: 1/86; the RSESE was not administered to HCs; the full distribution is provided in Supplementary Fig. S1). No participant scored ≥ 2, the conventional threshold for clinically significant EPS.

Inhibitory control, behavioral performance, and clinical correlates

Group differences in behavioral performance. Analyses of covariance (with age, sex, and years of education as covariates) revealed significant group differences across both tasks (Fig. 2A; Table 2). For the *Stroop* task, all conditions showed significant group differences (congruent, neutral, and incongruent; all $p < 0.01$), with the SCZ group consistently showing the slowest reaction times. The interference effect differed significantly across the groups ($F(2, 253) = 32.48$, $p < 0.001$, $\eta_p^2 = 0.204$), with the SCZ group exhibiting the largest interference relative to HCs ($p < 0.001$, $d = 1.52$), followed by the MDD group ($p = 0.002$, $d = 0.96$); the difference between the two patient groups was also significant ($p = 0.018$, $d = 0.66$). For the *Go/No-Go* task, significant group differences emerged for both Go and No-Go trials. Go accuracy ($F(2, 253) = 10.24$, $p < 0.001$) was significantly lower in SCZ than in HCs, whereas the MDD–HC difference did not reach significance ($p = 0.062$); Go reaction time ($F(2, 253) = 14.86$, $p < 0.001$) was prolonged in both patient groups relative to HCs. No-Go accuracy showed a significant graded pattern ($F(2, 253) = 28.64$, $p < 0.001$, $\eta_p^2 = 0.185$), with the SCZ group performing significantly worse than both HCs ($p < 0.001$, $d = 1.44$) and MDD ($p = 0.006$, $d = 0.63$), and the MDD group performing worse than HCs ($p = 0.008$, $d = 0.89$); a comparable graded pattern emerged for d' ($F(2, 253) = 34.26$, $p < 0.001$, $\eta_p^2 = 0.213$).

Measurement structure of inhibitory control. The two-factor structure of inhibitory control was preserved across the diagnostic groups. The two-factor model (interference inhibition and response inhibition) showed good fit in the full sample (CFI = 0.968,

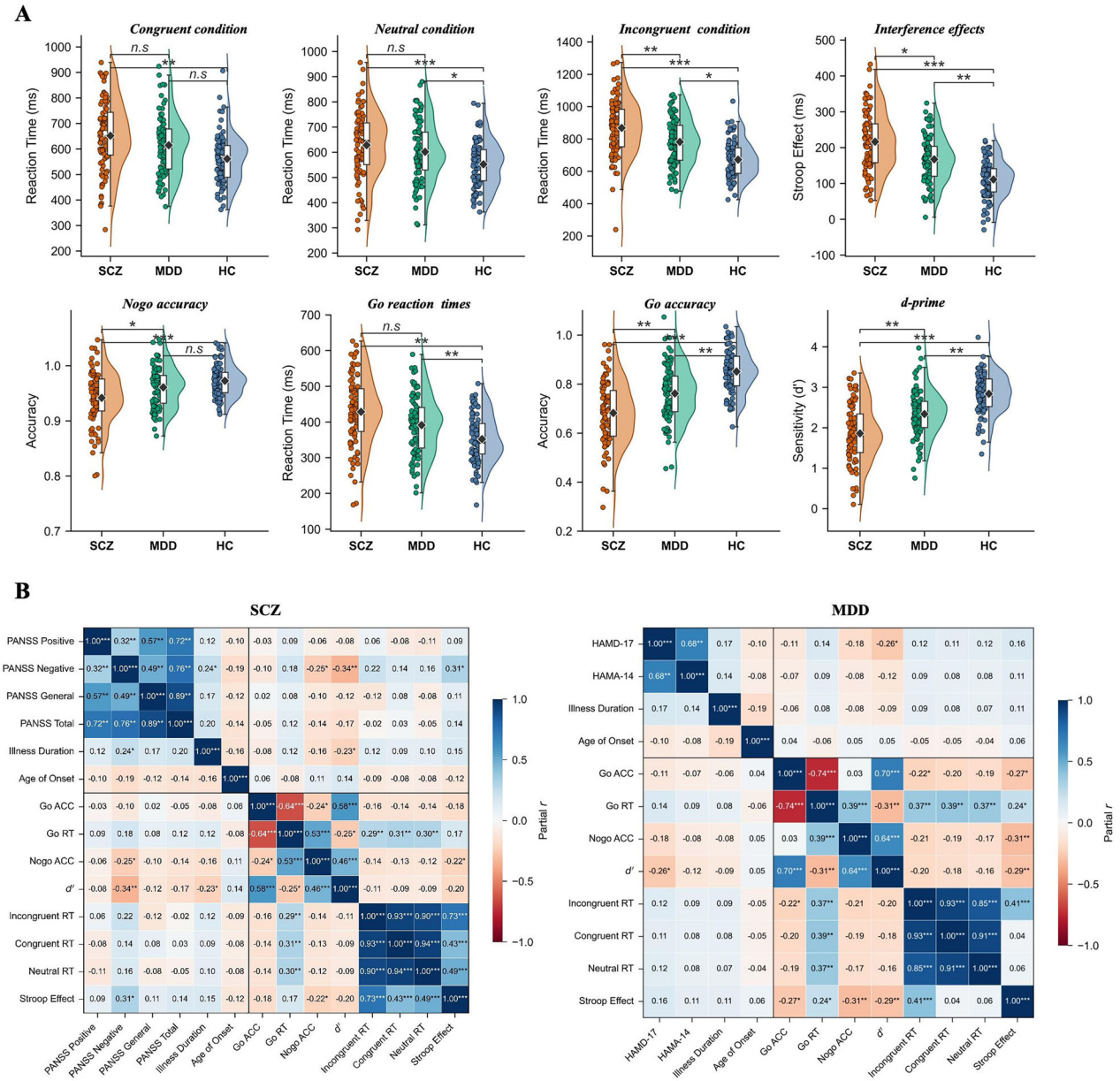


Fig. 2 Behavioral performance and clinical correlates of inhibitory control. A Raincloud plots depicting group differences in *Stroop* task indices (congruent, neutral, and incongruent reaction times and the interference effect) and *Go/No-Go* task indices (Go accuracy, Go reaction time, No-Go accuracy, and *d'*). Group comparisons were conducted using analysis of covariance with age, sex, and years of education as covariates. **B** Partial correlations between behavioral performance indices and clinical symptoms, controlling for age, sex, years of education, and medication dose (chlorpromazine equivalents for SCZ; fluoxetine equivalents for MDD). For both panels, *p* values were corrected using the Benjamini–Hochberg false discovery rate (FDR) procedure. ACC accuracy, Con congruent, CPZ chlorpromazine, *d'* discriminability index, FDR false discovery rate, FLU fluoxetine, HAMA-14 14-item Hamilton anxiety rating scale, HAMD-17 17-item Hamilton depression rating scale, HC healthy control, Inc incongruent, MDD major depressive disorder, Neu neutral, PANSS positive and negative syndrome scale, RT reaction time, SCZ schizophrenia. *$p_{FDR} < 0.05$, **$p_{FDR} < 0.01$, ***$p_{FDR} < 0.001$.

TLI = 0.956, RMSEA = 0.048 [90% CI: 0.028, 0.068], SRMR = 0.042) and acceptable fit within each group (Supplementary Fig. S2; Supplementary Table S8). Multi-group analysis supported configural invariance (CFI = 0.962), metric invariance (|ΔCFI| = 0.006), and partial scalar invariance (|ΔCFI| = 0.006) across the groups. Inter-factor correlations were positive in all three groups (HC: $r = 0.62$; MDD: $r = 0.48$; SCZ: $r = 0.38$), preserving the conceptual distinction between the two inhibitory subcomponents across diagnostic categories. A model additionally constraining the inter-factor covariance to be equal across groups yielded |ΔCFI| = 0.009, within the conventional 0.010 cutoff, supporting practical-level invariance of the two-factor measurement structure. The descriptively lower inter-factor correlation in SCZ ($r = 0.38$) than in HCs ($r = 0.62$), although within the practical-invariance threshold, warrants targeted replication.

Correlations between inhibitory control and clinical symptoms. Within-group partial correlations examined the associations between behavioral inhibitory control measures and clinical symptoms in each patient group, controlling for age, sex, years of education, and medication dose (chlorpromazine equivalents for SCZ; fluoxetine equivalents for MDD; Fig. 2B). In the SCZ group, PANSS-negative symptoms were significantly associated with both the *Stroop* interference effect ($r = 0.31$, $p_{FDR} = 0.018$) and *Go/No-*

Table 2. Inhibitory control task performance across the groups.

Measure	SCZ ($N = 87$)	MDD ($N = 86$)	HC ($N = 86$)	F-value	p_{FDR}-value	Tukey's HSD post hoc comparison		
						SCZ vs. MDD	SCZ vs. HCs	MDD vs. HCs
Stroop task								
Congruent RT (ms)	652.18 ± 138.64	614.72 ± 116.48	561.34 ± 96.82	6.42	**0**.002	0.118	**0**.001	0.052
Incongruent RT (ms)	868.42 ± 172.36	782.16 ± 142.58	672.48 ± 114.26	24.86	< **0**.001	**0**.004	< **0**.001	**0**.016
Neutral RT (ms)	628.36 ± 132.48	601.84 ± 118.62	552.16 ± 92.74	9.86	< **0**.001	0.274	< **0**.001	**0**.016
Stroop interference (ms)	216.24 ± 82.46	167.44 ± 64.28	111.14 ± 52.36	32.48	< **0**.001	**0**.018	< **0**.001	**0**.002
Go/No-Go task								
Go accuracy	0.94 ± 0.05	0.96 ± 0.04	0.97 ± 0.03	10.24	< **0**.001	**0**.042	< **0**.001	0.062
Go RT (ms)	428.64 ± 96.82	391.28 ± 82.46	352.18 ± 66.74	14.86	< **0**.001	0.184	< **0**.001	**0**.001
No-Go accuracy	0.68 ± 0.14	0.76 ± 0.11	0.85 ± 0.09	28.64	< **0**.001	**0**.006	< **0**.001	**0**.008
d'	1.86 ± 0.72	2.34 ± 0.58	2.84 ± 0.48	34.26	< **0**.001	**0**.006	< **0**.001	**0**.008

Data are mean ± *SD*. The three rightmost columns report Tukey's HSD post hoc *p*-values. Group differences were examined by ANCOVA with age, sex, and years of education as covariates; post hoc pairwise comparisons used Tukey's HSD test. Bold indicates statistical significance (FDR-corrected $p < 0.05$ for omnibus ANCOVA effects; Tukey-adjusted $p < 0.05$ for post hoc comparisons). *Stroop* interference = incongruent RT − congruent RT. d' was computed following standard signal-detection procedures, with Go trials treated as signal and No-Go trials as noise. Supplementary analyses indicated that these group-level differences were unlikely to be primarily attributable to differential practice-related acquisition across groups (Supplementary Table S5).

ANCOVA analysis of covariance, d' discriminability index, *HC* healthy control, *HSD* honestly significant difference, *MDD* major depressive disorder, *RT* reaction time, *SCZ* schizophrenia, *SD* standard deviation.

Go d' ($r = -0.34$, $p_{FDR} = 0.009$), and illness duration showed a significant negative association with d' ($r = -0.23$, $p_{FDR} = 0.046$). No significant associations emerged with PANSS-positive or PANSS-general measures (all $p_{FDR} > 0.05$). In the MDD group, the only FDR-significant behavioral–clinical association was a negative correlation between HAMD-17 total scores and d' ($r = -0.26$, $p_{FDR} = 0.024$); HAMA-14, illness duration, and age at onset did not reach significance (all $p_{FDR} > 0.05$). The corresponding partial correlations without medication-dose adjustment are reported in Supplementary Fig. S3A.

Computational mechanisms of inhibitory control impairment

Model fitting and diagnostics. HDDM provided a good fit for data from both tasks. All parameters had $\hat{R} < 1.01$, and MCMC trace plots indicated good mixing across chains for both the *Stroop* task (Supplementary Fig. S4) and the *Go/No-Go* task (Supplementary Fig. S5). Convergence statistics are summarized in Supplementary Table S9. Posterior predictive checks further demonstrated that simulated data closely matched observed data for both the *Stroop* (Supplementary Fig. S6B) and *Go/No-Go* (Supplementary Fig. S6A) tasks. Model comparison indicated that drift rate was the key parameter for distinguishing among the groups: the drift-rate regression model provided the best fit for both the *Stroop* and *Go/No-Go* tasks (Supplementary Table S10).

Group differences in HDDM parameters. Bayesian inference revealed substantial group differences in HDDM parameters (Fig. 3A). *(1) Drift rate*. Both patient groups showed substantially lower drift rate than HCs on both tasks. For the SCZ group, the posterior mean differences (PMDs) were −0.624 on the *Stroop* task (95% HDI [−0.862, −0.386]; $BF_{10} = 186.42$) and −0.548 on the *Go/No-Go* task (95% HDI [−0.782, −0.314]; $BF_{10} = 68.24$). For the MDD group, the PMDs were −0.386 on the *Stroop* task (95% HDI [−0.618, −0.154]; $BF_{10} = 16.86$) and −0.324 on the *Go/No-Go* task (95% HDI [−0.552, −0.096]; $BF_{10} = 8.42$). Direct comparisons yielded $BF_{10} = 3.86$ on the *Stroop* task and $BF_{10} = 2.96$ on the *Go/No-Go* task for a lower drift rate in SCZ relative to MDD. The SCZ drift-rate reduction remained robust after processing-speed adjustment (within-paradigm for *Stroop* and cross-paradigm for *Go/No-Go*; $BF_{10} = 38.7$–86.4; Supplementary Table S11). *(2) Non-decision time*. The SCZ group showed substantial T_{er} prolongation on both tasks (*Stroop*: PMD = 0.124, 95% HDI [0.068, 0.180], $BF_{10} = 12.86$; *Go/No-Go*: PMD = 0.108, 95% HDI [0.048, 0.168], $BF_{10} = 6.42$). In contrast, the MDD group showed weaker evidence for prolongation on the *Go/No-Go* task (PMD = 0.062, 95% HDI [0.008, 0.116], $BF_{10} = 2.24$) and no reliable effect on the *Stroop* task (PMD = 0.048, 95% HDI [−0.024, 0.120], $BF_{10} = 0.68$). Direct comparison yielded $BF_{10} = 4.68$ for longer T_{er} in SCZ relative to MDD on the *Stroop* task. Under processing-speed adjustment (within-paradigm for *Stroop* and cross-paradigm for *Go/No-Go*), the SCZ vs. HC T_{er} prolongation persisted on both tasks ($BF_{10} = 5.4$–7.8), whereas the MDD vs. HC T_{er} elevation on the *Go/No-Go* task did not survive adjustment ($BF_{10} = 1.6$; see Supplementary Results). *(3) Starting point bias*. On the *Go/No-Go* task, both patient groups showed lower values than HCs, indicating a weaker prior bias toward Go responses; evidence was moderate in SCZ (PMD = −0.068, 95% HDI [−0.098, −0.038], $BF_{10} = 6.24$) and only anecdotal in MDD (PMD = −0.042, 95% HDI [−0.072, −0.012], $BF_{10} = 2.18$). *(4) Decision boundary*. No substantial group differences were observed in decision boundary on either task (all $BF_{10} < 0.8$).

The principal SCZ drift-rate and T_{er} effects were further preserved under RPM-based fluid-intelligence adjustment ($\Delta\eta_p^2 = -0.027$ for *Stroop* and −0.024 for *Go/No-Go*; Supplementary Table S2) and after excluding the 7 RSESE-positive participants ($BF_{10} > 50$ for drift rate, > 5 for T_{er}; see Supplementary Results).

Parameter-decomposition analyses. Bootstrap parameter-decomposition analyses examined whether HDDM parameters accounted for the effect of task condition on behavioral performance within each group (Fig. 3B; Supplementary Fig. S7). Drift rate significantly accounted for condition effects on both the *Stroop* interference effect and d' across all three groups (all decomposed contributions: $p < 0.001$). For the *Stroop* task, the decomposed contributions of drift rate to the interference effect were 0.038 (95% CI [0.022, 0.056]) in SCZ and 0.028 (95% CI [0.014, 0.042]) in MDD. For the *Go/No-Go* task, the decomposed contributions of drift rate to d' were 0.054 (95% CI [0.036, 0.072]) in SCZ and 0.038 (95% CI [0.024, 0.054]) in MDD. Furthermore, starting point bias showed a significant decomposed contribution to d' in SCZ ($p = 0.018$), whereas the corresponding contribution in MDD did not reach significance ($p = 0.098$).

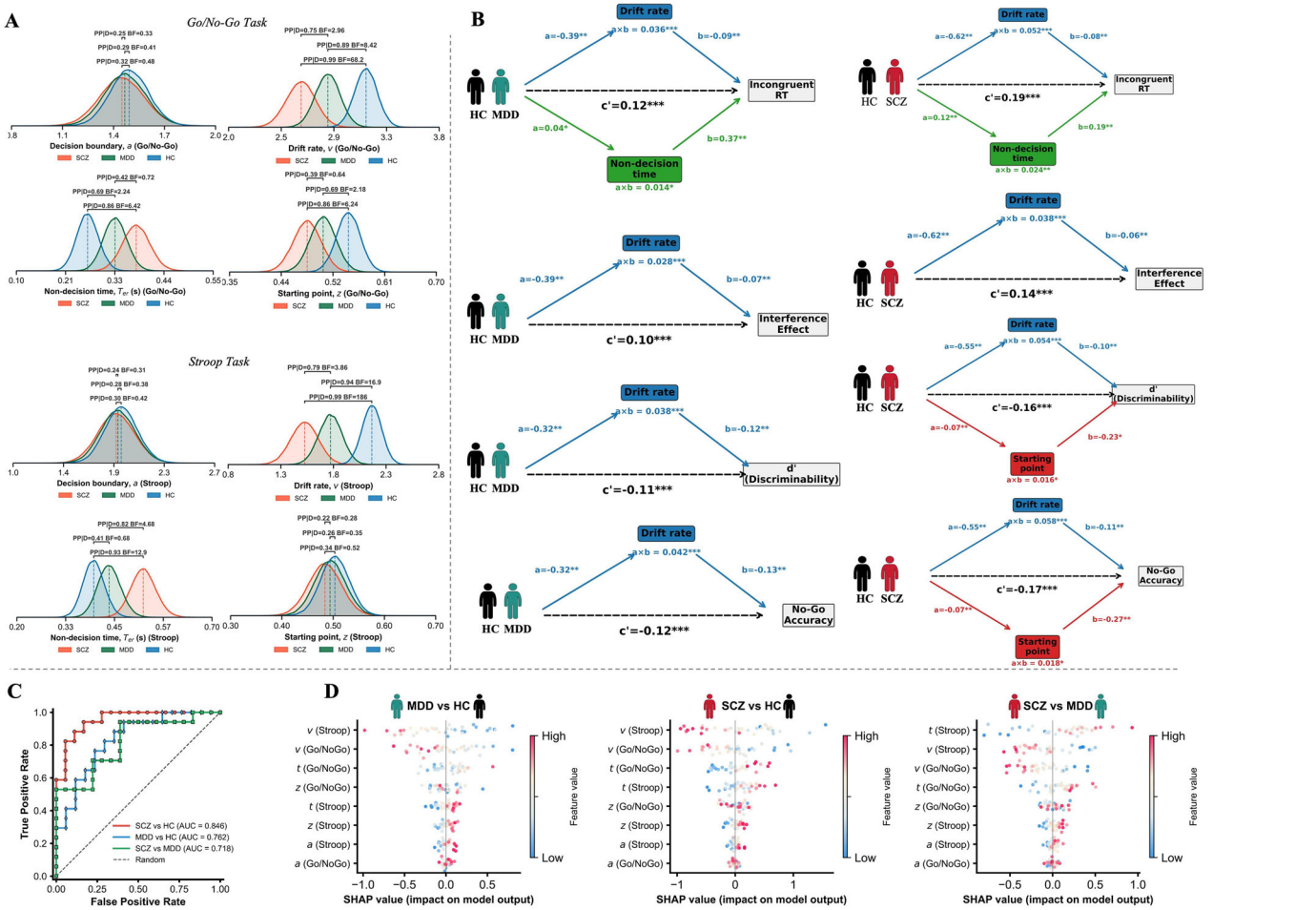


Fig. 3 HDDM parameter estimates, parameter-decomposition effects, and diagnostic classification. A Posterior distributions of HDDM parameters across the groups, shown separately for the *Go/No-Go* task (top) and *Stroop* task (bottom). Ridge plots display drift rate (*v*), non-decision time (T_{er}), and decision boundary (*a*) for each group (SCZ, MDD, and HCs in distinct colors), along with the starting point parameter (*z*) where applicable. Each panel's x-axis label includes the task name (e.g., 'Drift rate, *v* [*Stroop*]') for clarity. Vertical lines indicate posterior means, whereas shaded regions represent 95% highest density intervals. Brackets above each panel report posterior probabilities of group differences and Bayes factors (BF_{10}) for all three pairwise group comparisons, including non-substantial contrasts. **B** Parameter-decomposition analyses quantifying the contribution of HDDM parameters to within-group condition-to-performance effects. Path coefficients are standardized, and only decomposed contributions reaching significance are displayed; additional SCZ–MDD parameter-decomposition comparisons are reported in Supplementary Fig. S7. Because HDDM parameters are derived from the same RT and accuracy data being modeled, this analysis quantifies how HDDM decomposes condition-to-performance variance rather than constituting a causal mediation test. **C** Receiver operating characteristic (ROC) curves illustrating SVM classification performance using HDDM parameters as features. The three curves represent the pairwise diagnostic comparisons (SCZ vs. HCs, MDD vs. HCs, SCZ vs. MDD); AUC values are displayed in the legend. Results of the multiclass and demographic-confound sensitivity analyses are summarized in Supplementary Table S12; detailed multiclass diagnostics are shown in Supplementary Fig. S8. **D** SHAP summary plots illustrating feature importance for diagnostic classification. Features are ranked by mean absolute SHAP value (top to bottom), with each point representing one participant from the test set. Color indicates feature value (red and blue denote high and low values, respectively), and horizontal position indicates the feature's contribution to classification. *a* decision boundary, AUC area under the ROC curve, BF_{10} bayes factor, HC healthy control, HDDM hierarchical drift-diffusion model, MDD major depressive disorder, ROC receiver operating characteristic, SCZ schizophrenia, SHAP SHapley additive exPlanations, SVM support vector machine, T_{er} non-decision time, *v* drift rate, *z* starting point bias. *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$.

Diagnostic discrimination. SVM classifiers were used to evaluate the diagnostic discriminability of HDDM parameters (Fig. 3C). The classifier achieved good discrimination between SCZ and HCs (AUC = 0.846; BAC = 0.786), moderate discrimination between MDD and HCs (AUC = 0.762; BAC = 0.704), and above-chance discrimination between the two patient groups (AUC = 0.718; BAC = 0.684). In the two patient–control classifiers, SHAP analysis indicated that drift rate was the primary contributor to classification (mean |SHAP| range: 0.23–0.51), followed by non-decision time (0.09–0.34), whereas decision boundary contributed the least (< 0.06). By contrast, in the SCZ–MDD classifier, *Stroop* T_{er} was the top-ranked feature (mean |SHAP| = 0.34, versus 0.23 for drift rate; Fig. 3D), pointing to a dissociation between transdiagnostic and disorder-differentiating computational signatures: drift rate carried the dominant patient–control signal, whereas *Stroop* T_{er} carried the dominant SCZ–MDD signal.

As a direct comparison to the pairwise approach, a one-vs-rest multiclass SVM achieved a balanced accuracy of 0.67 [95% bootstrap CI: 0.55, 0.79] (permutation test $p < 0.001$), with per-class sensitivities of 0.77 (HCs), 0.65 (SCZ), and 0.61 (MDD). Detailed multiclass diagnostics, including receiver operating characteristic (ROC) curves and a confusion matrix, are provided in Supplementary Fig. S8. Demographic-confound sensitivity analyses further indicated that classification was driven predominantly by HDDM-derived computational information: within-group residualized pairwise performance attenuated only modestly

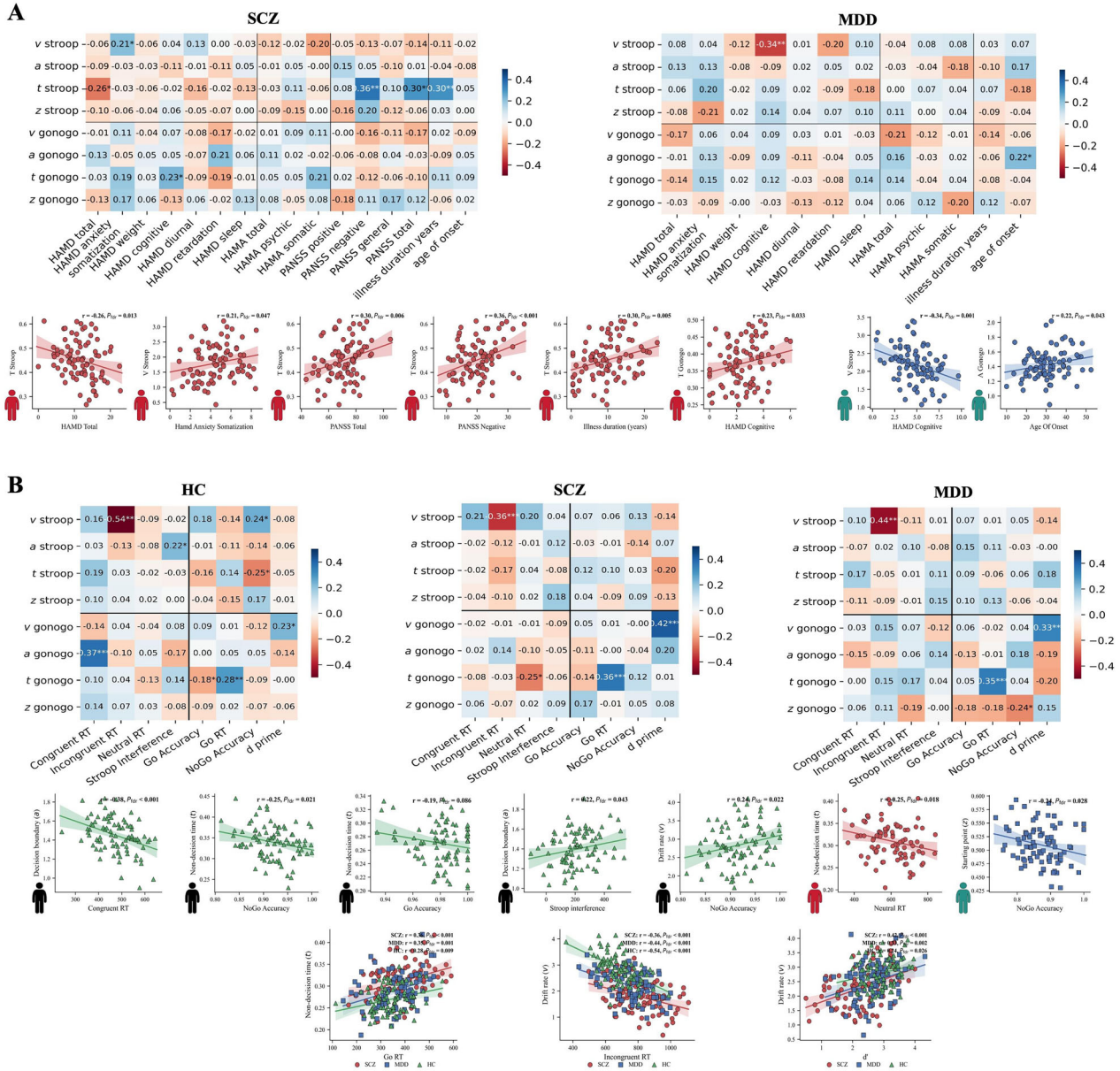


Fig. 4 Within-group partial correlations between HDDM parameters and clinical or behavioral indices. A Upper: Heatmaps displaying partial correlations between HDDM parameters and clinical symptom measures, shown separately for the SCZ group (left) and the MDD group (right), controlling for age, sex, years of education, and medication dose (chlorpromazine equivalents for SCZ; fluoxetine equivalents for MDD). Lower: Scatter plots with within-group regression lines depicting the FDR-significant cells in the upper heatmaps, shown separately for SCZ (red) and MDD (blue). **B** Upper: Heatmap displaying within-group partial correlations between HDDM parameters and behavioral inhibitory-control indices across all three groups, controlling for age, sex, and years of education. Lower: Scatter plots with within-group regression lines depicting the FDR-significant cells in the upper heatmap, displayed for all three groups (SCZ in red circles; MDD in blue squares; HCs in green triangles). All *p*-values were corrected using the Benjamini–Hochberg FDR procedure. *a* decision boundary, CPZ chlorpromazine, FDR false discovery rate, FLU fluoxetine, HAMA-14 14-item Hamilton anxiety rating scale, HAMD-17 17-item Hamilton depression rating scale, HC healthy control, HDDM hierarchical drift-diffusion model, MDD major depressive disorder, PANSS positive and negative syndrome scale, RT reaction time, SCZ schizophrenia, T_{er} non-decision time, *v* drift rate, *z* starting point bias. *$p_{FDR} < 0.05$, **$p_{FDR} < 0.01$, ***$p_{FDR} < 0.001$.

across the three contrasts ($\Delta = 0.017$–0.028), indicating that demographic and medication variance accounted for little of the primary classification performance (Supplementary Table S12).

Associations between HDDM parameters and behavioral and clinical indices

Across the groups, partial correlation analyses controlling for age, sex, and years of education revealed strong associations between HDDM parameters and behavioral inhibitory-control indices (Fig. 4B). Because HDDM parameters are estimated from the same trial-level RT and accuracy data, these associations index the internal coherence of the model decomposition rather than independent validation. Drift rate was significantly negatively correlated with the *Stroop* interference effect (SCZ: $r = -0.62$, $p_{FDR} < 0.001$; MDD: $r = -0.59$, $p_{FDR} < 0.001$; HC: $r = -0.47$, $p_{FDR} < 0.001$) and positively correlated with No-Go accuracy (SCZ: $r = 0.58$, $p_{FDR} < 0.001$; MDD: $r = 0.54$, $p_{FDR} < 0.001$; HC: $r = 0.49$, $p_{FDR} < 0.001$). Non-decision time was positively correlated with Go reaction time (SCZ: $r = 0.49$, $p_{FDR} < 0.001$; MDD: $r = 0.42$, $p_{FDR} < 0.001$; HC: $r = 0.46$, $p_{FDR} < 0.001$) but showed weaker associations

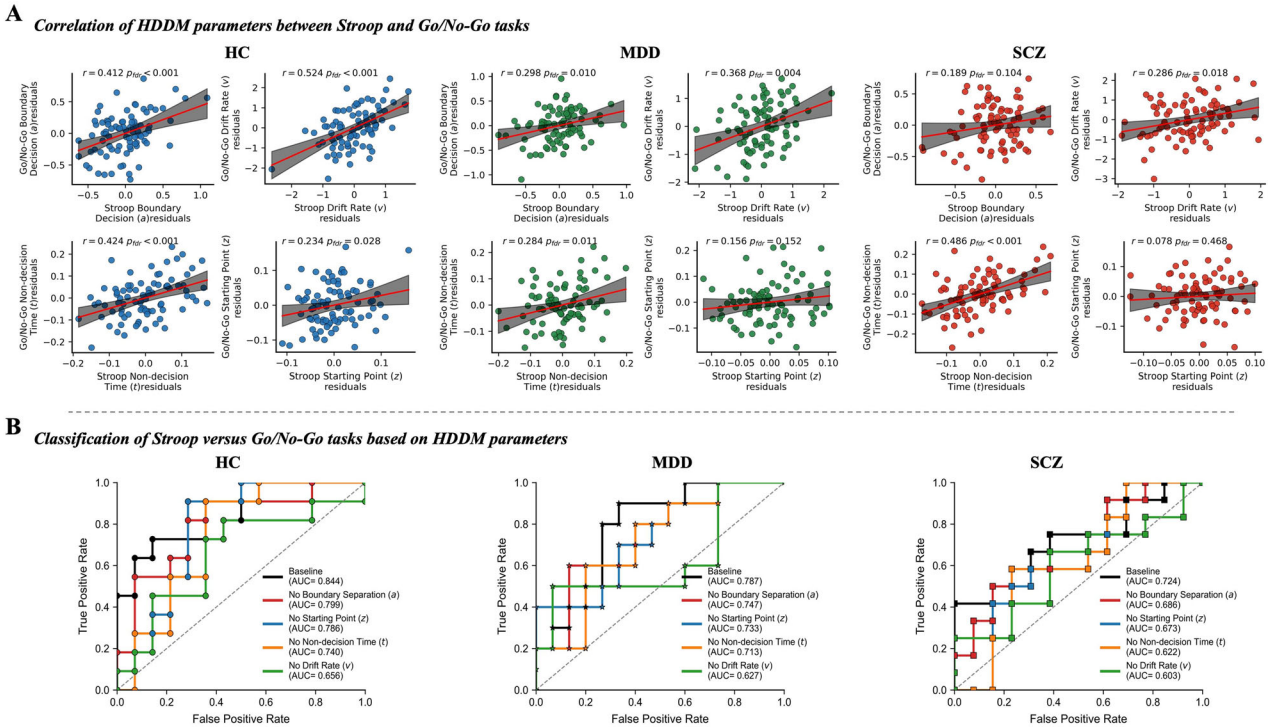


Fig. 5 Cross-task parameter consistency and task discriminability. A Within-subject correlations of HDDM parameter estimates between the *Stroop* and *Go/No-Go* tasks within each group. Each point represents one participant, and regression lines with 95% confidence intervals (CIs) are shown. **B** Receiver operating characteristic (ROC) curves for within-group SVM classification distinguishing the *Stroop* from the *Go/No-Go* task using the full HDDM parameter set (baseline) and feature-ablation models. Legends report the corresponding area under the curve (AUC) values. Full-model AUCs did not differ significantly across the groups (bootstrap difference tests, all $p > 0.15$). This task-type classification, in which each participant contributes one parameter set per task, is methodologically distinct from the individual-level diagnostic SVM (Fig. 3C; see Methods, Examination of shared and disorder-specific mechanisms). AUC area under the receiver operating characteristic curve, CI confidence interval, HC healthy control, HDDM hierarchical drift-diffusion model, MDD major depressive disorder, ROC receiver operating characteristic, SCZ schizophrenia, SVM support vector machine.

with the interference effect and No-Go accuracy. Starting point bias was positively correlated with Go accuracy (SCZ: $r = 0.37$, $p_{FDR} < 0.01$; MDD: $r = 0.31$, $p_{FDR} < 0.01$) and negatively correlated with No-Go accuracy (SCZ: $r = -0.30$, $p_{FDR} < 0.05$; MDD: $r = -0.26$, $p_{FDR} < 0.05$).

Analyses of clinical associations identified within-group symptom–parameter relationships in each diagnostic group (Fig. 4A). For these symptom–parameter associations, medication dose was additionally entered as a covariate (chlorpromazine equivalents for SCZ; fluoxetine equivalents for MDD), with FDR correction applied across the full set of HDDM parameter–symptom contrasts. In the SCZ group, *Stroop* T_{er} was positively correlated with PANSS-negative symptoms ($r = 0.36$, 95% CI [0.16, 0.53], $p_{FDR} < 0.01$) and with illness duration ($r = 0.30$, 95% CI [0.09, 0.48], $p_{FDR} < 0.05$). By contrast, the drift-rate association with PANSS-negative symptoms—significant before adjustment (*Stroop* $r = -0.37$)—attenuated to non-significance after medication-dose adjustment (*Stroop* $r = -0.13$, *Go/No-Go* $r = -0.16$; both $p_{FDR} > 0.05$), indicating that part of the drift–symptom covariance is carried by medication-related variance, whereas the T_{er}–negative symptom association remained robust. No FDR-significant associations emerged with PANSS-positive, PANSS-general, HAMD-17, or HAMA-14 measures (all $p_{FDR} > 0.05$). In the MDD group, the only FDR-significant clinical association was between the HAMD-17 cognitive factor and *Stroop* drift rate ($r = -0.34$, 95% CI [−0.52, −0.13], $p_{FDR} < 0.05$); HAMD-17 total, HAMA-14, illness duration, and age at onset showed no FDR-significant correlations. Medication-dose-adjusted correlations for the symptom measures displayed are provided in Supplementary Table S13. Partial correlations without medication-dose adjustment for the symptom–parameter contrasts are reported in Supplementary Fig. S3B.

Cross-task consistency and discriminability

Cross-task parameter consistency analysis showed that drift rate correlations between the two tasks were descriptively lower in the patient groups (SCZ: $r = 0.28$; MDD: $r = 0.42$) than in HCs ($r = 0.63$; Fig. 5A). Formal between-group testing using Fisher's *z*-transformation with Benjamini–Hochberg FDR correction across the three pairwise contrasts for each parameter showed that only the SCZ vs. HC drift-rate contrast was FDR-significant ($z = 2.93$, $p_{FDR} = 0.009$); all other pairwise drift-rate contrasts were non-significant (all $p_{FDR} > 0.12$). Cross-task T_{er} correlations showed no significant between-group differences (HC: $r = 0.42$; MDD: $r = 0.28$; SCZ: $r = 0.49$; all pairwise contrasts $p_{FDR} > 0.12$).

Task discriminability based on HDDM parameters was descriptively highest in HCs and lowest in SCZ, although full-feature AUCs did not differ significantly across the groups (bootstrap difference tests, all $p > 0.15$; Fig. 5B). In HCs, the AUC for differentiating the two tasks was 0.844 (permutation test $p < 0.001$), with drift rate as the strongest contributor ($\Delta AUC = -0.188$ upon removal). The SCZ group showed descriptively lower task discriminability ($AUC = 0.724$; permutation test $p = 0.008$), accompanied by an attenuated discriminative contribution of drift rate ($\Delta AUC = -0.121$). The MDD group showed a descriptively intermediate value ($AUC = 0.787$; permutation test $p = 0.048$), with drift rate remaining the primary discriminating parameter ($\Delta AUC = -0.160$).

DISCUSSION

The present study used HDDM to systematically examine the shared and distinct computational mechanisms underlying interference- and response-inhibition impairments in SCZ and MDD. At

the behavioral level, both patient groups exhibited significant deficits across most measures, with SCZ showing more severe impairment. At the computational level, reduced drift rate emerged as the core shared impairment, whereas non-decision time prolongation was disorder-differentiating—consistent across both tasks in SCZ but, in MDD, confined to response inhibition and not robust to processing-speed adjustment.

Behavioral patterns of interference and response inhibition impairments

The graded behavioral pattern from HCs to MDD and then to SCZ, with SCZ exhibiting the most severe impairment [3, 76], was paralleled at the parameter level: drift rate showed a descriptive cross-group gradient (HC > MDD > SCZ), and T_{er} showed disorder-specific prolongation that was robust in SCZ. Within-group parameter-decomposition analyses indicated that drift rate showed a significant decomposed contribution to the condition-to-performance relationship in each diagnostic group, with a smaller but significant decomposed contribution from starting point bias in SCZ. This convergence is consistent with the cross-group behavioral gradient and the within-group condition effects engaging a common evidence-accumulation stage, with disorder-specific T_{er} prolongation superimposed on this shared architecture in SCZ, although the present design cannot establish this directly. The intermediate position of MDD in the drift-rate gradient is consistent with a graded computational account in which inhibitory-control demands engage a partial form of the evidence-accumulation deficit, which is more fully expressed in SCZ.

The present cross-sectional data cannot adjudicate whether this gradient reflects illness-severity continuity, convergent cognitive sequelae, or partially shared etiology [8], nor does the design license inference about underlying neural substrates. Although SCZ is frequently conceptualized along a neurodevelopmental continuum [77], the present design cannot determine how that framework relates to the MDD-intermediate pattern. We therefore frame the graded pattern as hypothesis-generating, pending designs that compare severity-matched subsamples or longitudinal trajectories. Motivated by transdiagnostic associations between cognitive dysfunction and symptom dimensions [78], this graded account generates a falsifiable prediction: if the MDD-intermediate position reflects illness-severity continuity rather than diagnostically separate mechanisms, then a transdiagnostic severity dimension derived from instruments administered identically in both groups should yield a continuous inverse correlation with drift rate that crosses diagnostic boundaries, with no abrupt MDD–SCZ discontinuity.

Beyond the cross-group gradient, we also examined whether the two-factor structure of inhibitory control was preserved across the diagnostic groups. Multi-group CFA supported configural, metric, and partial scalar invariance of the two-factor structure (interference inhibition vs. response inhibition) in all three groups, consistent with multi-component theories of inhibitory control [18]. Although inter-factor correlations were descriptively lower in the patient groups than in HCs, constraining the inter-factor covariance to equality across groups yielded $|\Delta CFI| = 0.009$, within the conventional 0.010 cutoff [65], indicating that the measurement structure is practically invariant across the groups. We therefore frame the descriptively lower inter-factor coupling in the patient groups as hypothesis-generating rather than as an inferential finding. Because group invariance was met only at the practical-significance threshold, the apparent group difference in inter-factor coupling cannot be cleanly separated from residual differences in the measurement structure itself; targeted replication in independent samples will be needed to determine whether the pattern reflects genuine differences in the organization of inhibitory control or measurement-level artifacts.

Reduced drift rate in inhibitory control as a shared computational deficit

Consistent with our hypothesis, HDDM analysis identified reduced drift rate as the most robust computational impairment underlying inhibitory control deficits in both disorders, with patients with SCZ and MDD showing substantially lower drift rates than HCs on both tasks. This finding aligns with prior reports of reduced evidence accumulation in both disorders [31, 34], suggesting that inhibitory control deficits in both conditions may be rooted in a fundamental reduction in information-processing efficiency [3, 31, 33, 34], rather than in the selective impairment of any particular inhibitory mechanism. The SCZ drift-rate reduction was descriptively greater than that in MDD, paralleling the graded behavioral pattern [76].

The transdiagnostic nature of the drift-rate deficit raises the hypothesis of partly shared prefrontal mechanisms. Interference inhibition consistently engages the dorsal anterior cingulate cortex together with lateral prefrontal regions, which jointly support conflict detection, control allocation, and attentional biasing [19], whereas response inhibition consistently engages the right-lateralized fronto-striatal network that includes the right inferior frontal gyrus, pre-supplementary motor area, and basal ganglia [13, 29]. These overlapping control networks provide a plausible—but untested—circuit-level framework for the transdiagnostic drift-rate deficit [13, 19, 29]. At the neurotransmitter level, dopamine-system dysregulation has been implicated in both SCZ and depression [9, 20], whereas MDD also shows reduced reward responsiveness [79]. Models of basal-ganglia and prefrontal dopamine function provide a theoretical basis for a link to task-relevant signal-to-noise [80, 81], but this mechanism was not tested here. We frame the mapping between HDDM parameters and these neural circuits as a hypothesis for future neuroimaging studies rather than a claim derived from the present behavioral data.

Turning to the disorder-specific predictions, the SCZ drift-rate reduction was supported, whereas the candidate account of an elevated decision boundary in MDD was not corroborated, as Bayesian comparison provided no substantial evidence for decision-boundary differences across the groups. The lack of evidence for an elevated decision boundary in MDD is consistent with the possibility that heightened risk aversion reported specifically in depressed patients with a history of suicide attempts [82] may not generalize to inhibitory-control contexts, or that such effects may be more readily detected in reward-related decision-making paradigms than in cognitive control tasks; because the present analyses were not designed to provide positive evidence for equivalence, this null result should be interpreted with corresponding caution. Therefore, the principal cross-disorder dissociation in the present data lies not in decision-strategy parameters but in non-decision time.

Disorder-specific profiles in non-decision time and starting point bias

Beyond the shared drift-rate impairment, the two disorders diverged in their T_{er} profiles. Although psychomotor slowing is a well-established feature of both SCZ and MDD [83–86], the SCZ T_{er} prolongation was consistent across both tasks and persisted after adjustment for medication dose and within-paradigm processing-speed proxies. The SCZ T_{er} signal does not appear fully attributable to within-paradigm RT variance, although full dissociation from general processing speed awaits independent neuropsychological measurement (e.g., the *Digit Symbol Substitution* Test and *Trail Making Test Part* A). By contrast, the MDD T_{er} elevation on the *Go/No-Go* task did not survive the same conservative adjustment. Given that MDD is itself associated with psychomotor retardation [83, 84], we do not interpret this attenuation as absence of slowing per se, but as an indication that the MDD T_{er} signal is not robustly distinguishable from basic processing-speed variance under the

proxy used here. Under currently available within-paradigm proxies, the SCZ T_{er} signature is therefore more robust than the MDD T_{er} signature, with neither cleanly separable from basic processing speed in the absence of independent neuropsychological measurement.

The task-specific T_{er} prolongation observed in MDD on the *Go/No-Go* task may reflect differences in the response demands of the two paradigms rather than a generalized perceptual–motor deficit consistently expressed across paradigms. On the *Go/No-Go* task, starting point bias was reduced in both patient groups—most reliably in SCZ—indicating weakened prior preparation for Go responses. Rather than reflecting the more conservative decision strategy considered as one candidate account, this reduced bias is more consistent with impaired sensitivity to environmental statistical regularities [87–89], although this interpretation requires direct testing.

At the clinical level, computational parameters showed diverging symptom correlates in SCZ. After medication-dose adjustment, the *Stroop* T_{er}–negative symptom association persisted, whereas drift-rate associations with negative symptoms did not, suggesting that the perceptual–motor stage of decision processing carries a comparatively robust symptom-related signal in SCZ. The direction of the T_{er}–negative symptom relationship is descriptively consistent with established psychomotor-slowing accounts of negative symptomatology in schizophrenia [85]. In MDD, the surviving HAMD-17 cognitive factor–*Stroop* drift-rate association indicates that the depressive symptom–parameter signal localizes specifically to the interference paradigm under medication-adjusted analyses, rather than reflecting a generalized symptom–performance covariation. The attenuation of several originally significant associations below the FDR threshold under medication-dose adjustment underscores that part of the cross-sectional symptom–parameter covariance in psychiatric samples may be confounded with medication-related variance, and highlights the value of reporting both adjusted and unadjusted symptom–parameter associations in computational-psychiatry studies of medicated populations.

Cross-task parameter consistency and task discriminability

We examined cross-task parameter consistency to assess the within-individual organization of the inhibitory control system. Cross-task drift-rate correlations were descriptively lower in the patient groups than in HCs, with formal between-group testing revealing that only the SCZ vs. HC contrast survived FDR correction. Cross-task T_{er} correlations did not differ significantly between groups. Across the HDDM parameters tested, only the SCZ vs. HC drift-rate contrast survived correction, indicating that cross-task consistency in evidence accumulation is reduced in SCZ relative to HCs, although the SCZ–MDD contrast did not itself survive correction; prior neuroimaging work has linked inter-individual differences in cognitive performance in SCZ to heterogeneous prefrontal dysconnectivity [90], although that neural heterogeneity is not equivalent to the cross-task behavioral consistency measured here.

In parallel, task discriminability based on HDDM parameters was descriptively highest in HCs and lowest in SCZ, although full-feature AUCs did not differ significantly across the groups, with drift rate emerging as the strongest discriminating parameter across all groups. This task-type classification, in which each participant contributes one parameter set per task and the classification target is task type, is methodologically distinct from the individual-level diagnostic SVM reported earlier (Fig. 3C), in which the classification target is diagnostic group. Together, these results suggest a descriptive pattern in which the cross-task computational fingerprint is most consistent in HCs and least consistent in SCZ, with MDD occupying an intermediate position—a pattern that warrants targeted replication in future work.

Limitations and future directions

Several limitations warrant consideration. First, measurement and design factors constrain inferences about cross-disorder specificity. The cross-sectional design cannot establish whether reduced drift rate is a cause, consequence, or concomitant of illness; longitudinal tracking in first-episode or high-risk individuals will be needed to adjudicate [91]. We also used diagnosis-specific symptom instruments (PANSS for SCZ; HAMD-17 for MDD) rather than parallel transdiagnostic batteries, an approach also used in prior cross-disorder studies [38, 92, 93]. The absence of the Calgary Depression Scale for Schizophrenia (CDSS) is particularly consequential for the SCZ group, in which depressive symptomatology cannot be cleanly separated from negative symptoms and antipsychotic-related motor effects [94, 95]; because all patients were medicated at assessment, retrospective supplementation would have been confounded by medication status and temporal misalignment with the cognitive battery. Within-group symptom–parameter associations are therefore reported as hypothesis-generating, whereas the core task-derived findings rest on tasks administered identically across all groups and are unaffected by this asymmetry. Cognitive assessment was likewise limited to RPM and the two inhibitory tasks, omitting processing speed, working memory, declarative memory, selective attention, and premorbid IQ; given the documented premorbid intellectual decline in SCZ [96, 97], premorbid-IQ proxies (e.g., the Wechsler Test of Adult Reading or the Wide Range Achievement Test–Reading subtest) and symmetric transdiagnostic batteries (e.g., the CDSS with the Brief Negative Symptom Scale or the Clinical Assessment Interview for Negative Symptoms) should be a priority for replication.

Second, sample and task characteristics bound generalizability. The near-parity sex distribution across groups departs from the population-level male predominance in SCZ and female predominance in MDD [98, 99]; while this reduced sex as a confound, replication in epidemiologically representative samples with formal sex-moderation analyses will strengthen generalizability. The emotionally neutral tasks isolated cold cognitive control but may have understated affect-laden inhibitory deficits in MDD, and inhibitory control includes further components (e.g., cognitive inhibition) not assessed here [12]. Future work using a broader paradigm set—*stop-signal* [100], directed forgetting, negative priming, and emotional *Stroop* variants with depression-relevant material [101, 102]—could test whether reduced drift rate generalizes across subdomains and whether MDD parameter abnormalities are amplified under affective demands. The protocol also did not log practice-block attempt counts, precluding attempts-to-criterion analyses; future studies should preregister trial-by-trial practice logs.

Third, residual medication and processing-speed confounds remain despite our multi-layered sensitivity analyses (medication-dose adjustment, within-paradigm processing-speed proxy, RPM adjustment, and RSESE-positive case exclusion), all of which preserved the principal SCZ drift-rate reduction and SCZ vs. HC T_{er} elevation. Because the within-paradigm proxy itself partially indexes T_{er}, the more robust SCZ T_{er} signal should not be read as fully dissociable from general processing speed without an independent neuropsychological measure. EPS positivity was near floor across groups, consistent with EPS not being the primary driver of T_{er} prolongation; however, this does not exclude subclinical motor signs undetectable by the binary RSESE indicator, nor non-motor pharmacological effects (e.g., sedation, anticholinergic burden) that may influence T_{er} [103, 104]. Future medicated–unmedicated or longitudinal pre-/post-medication designs, with more sensitive motor assessments (the Extrapyramidal Symptom Rating Scale, the Abnormal Involuntary Movement Scale, or accelerometry), would better separate disease from medication effects [105].

Fourth, the absence of neuroimaging precluded direct examination of how computational parameters map onto brain function. Although drift rate has been linked to prefrontal–striatal circuits and non-decision time to sensorimotor activity [106], whether these associations hold in psychiatric populations remains open; functional MRI or EEG could probe the neural basis of HDDM parameters and test whether computational impairments mediate links between neural and behavioral abnormalities [107].

In summary, this study delineated the computational mechanisms underlying inhibitory control impairments in SCZ and MDD. Reduced drift rate emerged as a shared transdiagnostic deficit in evidence-accumulation efficiency, being descriptively more pronounced in SCZ, whereas T_{er} prolongation followed a disorder-differentiating pattern—consistent across both tasks in SCZ but, in MDD, confined to response inhibition and not robust to processing-speed adjustment—with definitive dissociation from general processing speed awaiting independent neuropsychological measurement. Within each diagnostic group, computational parameters showed exploratory cross-sectional symptom correlates: prolonged *Stroop* T_{er} was associated with negative-symptom severity in SCZ, and reduced *Stroop* drift rate with depressive cognitive symptoms in MDD. Pending prospective replication with independent processing-speed measures and longitudinal designs, computational phenotyping of inhibitory control could provide a quantitative adjunct to symptom-based assessment when SCZ–MDD diagnostic overlap is clinically ambiguous.

DATA AVAILABILITY

Data are available at https://github.com/tyzhang98/Inhibitory-Control-SCZ-MDD-Data.

CODE AVAILABILITY

Scripts for the main analyses are available at https://github.com/tyzhang98/Inhibitory-Control-SCZ-MDD-Code.

REFERENCES

1. Rock PL, Roiser JP, Riedel WJ, Blackwell AD. Cognitive impairment in depression: a systematic review and meta-analysis. Psychol Med. 2014;44:2029–40.
2. Schaefer J, Giangrande E, Weinberger DR, Dickinson D. The global cognitive impairment in schizophrenia: consistent over decades and around the world. Schizophr Res. 2013;150:42–50.
3. Snyder HR. Major depressive disorder is associated with broad impairments on neuropsychological measures of executive function: a meta-analysis and review. Psychol Bull. 2013;139:81–132.
4. Langenecker SA, Kennedy SE, Guidotti LM, Briceno EM, Own LS, Hooven T, et al. Frontal and limbic activation during inhibitory control predicts treatment response in major depressive disorder. Biol Psychiatry. 2007;62:1272–80.
5. Cuthbert BN, Insel TR. Toward the future of psychiatric diagnosis: the seven pillars of RDoC. BMC Med. 2013;11:126.
6. Insel T, Cuthbert B, Garvey M, Heinssen R, Pine DS, Quinn K, et al. Research domain criteria (RDoC): toward a new classification framework for research on mental disorders. Am J Psychiatry. 2010;167:748–51.
7. Brosch K, Stein F, Schmitt S, Pfarr J-K, Ringwald KG, Thomas-Odenthal F, et al. Reduced hippocampal gray matter volume is a common feature of patients with major depression, bipolar disorder, and schizophrenia spectrum disorders. Mol Psychiatry. 2022;27:4234–43.
8. Cross-Disorder Group of the Psychiatric Genomics Consortium. Identification of risk loci with shared effects on five major psychiatric disorders: a genome-wide analysis. Lancet. 2013;381:1371–9.
9. Grace AA. Dysregulation of the dopamine system in the pathophysiology of schizophrenia and depression. Nat Rev Neurosci. 2016;17:524–32.
10. Kirschner H, Nassar MR, Fischer AG, Frodl T, Meyer-Lotz G, Froböse S, et al. Transdiagnostic inflexible learning dynamics explain deficits in depression and schizophrenia. Brain. 2024;147:201–14.
11. Badcock JC, Hugdahl K. A synthesis of evidence on inhibitory control and auditory hallucinations based on the Research Domain Criteria (RDoC) framework. Front Hum Neurosci. 2014;8:180.
12. Nigg JT. On inhibition/disinhibition in developmental psychopathology: views from cognitive and personality psychology and a working inhibition taxonomy. Psychol Bull. 2000;126:220–46.
13. Aron AR, Robbins TW, Poldrack RA. Inhibition and the right inferior frontal cortex: one decade on. Trends Cogn Sci. 2014;18:177–85.
14. Diamond A. Executive functions. Annu Rev Psychol. 2013;64:135–68.
15. Lipszyc J, Schachar R. Inhibitory control and psychopathology: a meta-analysis of studies using the stop signal task. J Int Neuropsychol Soc. 2010;16:1064–76.
16. Westerhausen R, Kompus K, Hugdahl K. Impaired cognitive inhibition in schizophrenia: a meta-analysis of the Stroop interference effect. Schizophr Res. 2011;133:172–81.
17. Weisbrod M, Kiefer M, Marzinzik F, Spitzer M. Executive control is disturbed in schizophrenia: evidence from event-related potentials in a Go/NoGo task. Biol Psychiatry. 2000;47:51–60.
18. Friedman NP, Miyake A. The relations among inhibition and interference control functions: a latent-variable analysis. J Exp Psychol Gen. 2004;133:101–35.
19. Botvinick MM, Braver TS, Barch DM, Carter CS, Cohen JD. Conflict monitoring and cognitive control. Psychol Rev. 2001;108:624–52.
20. Howes OD, Kapur S. The dopamine hypothesis of schizophrenia: version III—the final common pathway. Schizophr Bull. 2009;35:549–62.
21. Hughes ME, Fulham WR, Johnston PJ, Michie PT. Stop-signal response inhibition in schizophrenia: behavioural, event-related potential and functional neuroimaging data. Biol Psychol. 2012;89:220–31.
22. Zandbelt BB, van Buuren M, Kahn RS, Vink M. Reduced proactive inhibition in schizophrenia is related to corticostriatal dysfunction and poor working memory. Biol Psychiatry. 2011;70:1151–8.
23. Joormann J, Gotlib IH. Emotion regulation in depression: relation to cognitive inhibition. Cogn Emot. 2010;24:281–98.
24. Clawson A, Clayson PE, Larson MJ. Cognitive control adjustments and conflict adaptation in major depressive disorder. Psychophysiology. 2013;50:711–21.
25. Vanderhasselt M-A, De Raedt R, De Paepe A, Aarts K, Otte G, Van Dorpe J, et al. Abnormal proactive and reactive cognitive control during conflict processing in major depression. J Abnorm Psychol. 2014;123:68–80.
26. Ratcliff R, McKoon G. The diffusion decision model: theory and data for two-choice decision tasks. Neural Comput. 2008;20:873–922.
27. Wiecki TV, Sofer I, Frank MJ. HDDM: hierarchical Bayesian estimation of the drift-diffusion model in Python. Front Neuroinform. 2013;7:14.
28. Pan W, Geng H, Zhang L, Fengler A, Frank MJ, Zhang R-Y, et al. dockerHDDM: a user-friendly environment for Bayesian hierarchical drift-diffusion modeling. Adv Methods Pract Psychol Sci. 2025;8:25152459241298700.
29. Simmonds DJ, Pekar JJ, Mostofsky SH. Meta-analysis of Go/No-go tasks demonstrating that fMRI activation associated with response inhibition is task-dependent. Neuropsychologia. 2008;46:224–32.
30. Swick D, Ashley V, Turken U. Are the neural correlates of stopping and not going identical? Quantitative meta-analysis of two response inhibition tasks. Neuroimage. 2011;56:1655–65.
31. Fish S, Toumaian M, Pappa E, Davies TJ, Tanti R, Saville CWN, et al. Modelling reaction time distribution of fast decision tasks in schizophrenia: evidence for novel candidate endophenotypes. Psychiatry Res. 2018;269:212–20.
32. Mathias SR, Knowles EEM, Barrett J, Leach O, Buccheri S, Beetham T, et al. The processing-speed impairment in psychosis is more than just accelerated aging. Schizophr Bull. 2017;43:814–23.
33. Dickinson D, Ramsey ME, Gold JM. Overlooking the obvious: a meta-analytic comparison of digit symbol coding tasks and other cognitive measures in schizophrenia. Arch Gen Psychiatry. 2007;64:532–42.
34. Lawlor VM, Webb CA, Wiecki TV, Frank MJ, Trivedi M, Pizzagalli DA, et al. Dissecting the impact of depression on decision-making. Psychol Med. 2020;50:1613–22.
35. Pe ML, Vandekerckhove J, Kuppens P. A diffusion model account of the relationship between the emotional flanker task and rumination and depression. Emotion. 2013;13:739–47.
36. Dillon DG, Lazarov A, Dolan S, Bar-Haim Y, Pizzagalli DA, Schneier FR. Fast evidence accumulation in social anxiety disorder enhances decision making in a probabilistic reward task. Emotion. 2022;22:1–18.
37. Moustafa AA, Kéri S, Somlai Z, Balsdon T, Frydecka D, Misiak B, et al. Drift diffusion model of reward and punishment learning in schizophrenia: modeling and experimental data. Behav Brain Res. 2015;291:147–54.
38. Shen C, Calvin OL, Rawls E, Redish AD, Sponheim SR. Clarifying cognitive control deficits in psychosis via drift diffusion modeling and attractor dynamics. Schizophr Bull. 2024;50:1357–70.
39. Huang-Pollock C, Ratcliff R, McKoon G, Shapiro Z, Weigard A, Galloway-Long H. Using the diffusion model to explain cognitive deficits in attention deficit hyperactivity disorder. J Abnorm Child Psychol. 2017;45:57–68.
40. Gupta A, Bansal R, Alashwal H, Kacar AS, Balci F, Moustafa AA. Neural substrates of the drift-diffusion model in brain disorders. Front Comput Neurosci. 2022;15:678232.

41. Gaillard A, Tan EJ, Carruthers SP, Gurvich C, Hughes ME, Neill E, et al. No influence of sex on the relationship between schizotypy factors and executive control across the schizophrenia spectrum. J Psychiatr Res. 2022;148:325–31.
42. Aleman A, Kahn RS, Selten J-P. Sex differences in the risk of schizophrenia: evidence from meta-analysis. Arch Gen Psychiatry. 2003;60:565–71.
43. Salk RH, Hyde JS, Abramson LY. Gender differences in depression in representative national samples: meta-analyses of diagnoses and symptoms. Psychol Bull. 2017;143:783–822.
44. Faul F, Erdfelder E, Lang A-G, Buchner A. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. Behav Res Methods. 2007;39:175–91.
45. First MB, Spitzer RL, Gibbon M, Williams JBW Structured clinical interview for DSM-IV Axis I disorders—patient edition (SCID-I/P, version 20, 4/97 revision). New York: Biometrics Research Department, New York State Psychiatric Institute; 1997.
46. Zhang T, Zhao X, Yeo BTT, Huo X, Eickhoff SB, Chen J. Leveraging stacked classifiers for multi-task executive function in schizophrenia yields diagnostic and prognostic insights. Schizophr Bull. 2026;52:sbaf218.
47. Sheehan DV, Lecrubier Y, Sheehan KH, Amorim P, Janavs J, Weiller E, et al. The Mini-International Neuropsychiatric Interview (M.I.N.I.): the development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. J Clin Psychiatry. 1998;59:22–33. Suppl 20.
48. Bradley RH, Corwyn RF. Socioeconomic status and child development. Annu Rev Psychol. 2002;53:371–99.
49. Hackman DA, Farah MJ. Socioeconomic status and the developing brain. Trends Cogn Sci. 2009;13:65–73.
50. Achim AM, Maziade M, Raymond É, Olivier D, Mérette C, Roy M-A. How prevalent are anxiety disorders in schizophrenia? A meta-analysis and critical review on a significant association. Schizophr Bull. 2011;37:811–21.
51. Hamilton M. A rating scale for depression. J Neurol Neurosurg Psychiatry. 1960;23:56–62.
52. Hamilton M. The assessment of anxiety states by rating. Br J Med Psychol. 1959;32:50–55.
53. Kay SR, Fiszbein A, Opler LA. The positive and negative syndrome scale (PANSS) for schizophrenia. Schizophr Bull. 1987;13:261–76.
54. Leucht S, Samara M, Heres S, Davis JM. Dose equivalents for antipsychotic drugs: the DDD method. Schizophr Bull. 2016;42:S90–S94.
55. Hayasaka Y, Purgato M, Magni LR, Ogawa Y, Takeshima N, Cipriani A, et al. Dose equivalents of antidepressants: evidence-based recommendations from randomized controlled trials. J Affect Disord. 2015;180:179–84.
56. Simpson GM, Angus JWS. A rating scale for extrapyramidal side effects. Acta Psychiatr Scand Suppl. 1970;212:11–19.
57. Laird AR, McMillan KM, Lancaster JL, Kochunov P, Turkeltaub PE, Pardo JV, et al. A comparison of label-based review and ALE meta-analysis in the Stroop task. Hum Brain Mapp. 2005;25:6–21.
58. Roiser JP, Sahakian BJ. Hot and cold cognition in depression. CNS Spectr. 2013;18:139–49.
59. Salehinejad MA, Ghanavati E, Rashid MHA, Nitsche MA. Hot and cold executive functions in the brain: a prefrontal-cingular network. Brain Neurosci Adv. 2021;5:23982128211007769.
60. Stroop JR. Studies of interference in serial verbal reactions. J Exp Psychol. 1935;18:643–62.
61. Stanislaw H, Todorov N. Calculation of signal detection theory measures. Behav Res Methods Instrum Comput. 1999;31:137–49.
62. Dutilh G, Vandekerckhove J, Tuerlinckx F, Wagenmakers E-J. A diffusion model decomposition of the practice effect. Psychon Bull Rev. 2009;16:1026–36.
63. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc Ser B Stat Methodol. 1995;57:289–300.
64. Karr JE, Areshenkoff CN, Rast P, Hofer SM, Iverson GL, Garcia-Barrera MA. The unity and diversity of executive functions: a systematic review and re-analysis of latent variable studies. Psychol Bull. 2018;144:1147–85.
65. Cheung GW, Rensvold RB. Evaluating goodness-of-fit indexes for testing measurement invariance. Struct Equ Modeling. 2002;9:233–55.
66. Vandenberg RJ, Lance CE. A review and synthesis of the measurement invariance literature: suggestions, practices, and recommendations for organizational research. Organ Res Methods. 2000;3:4–70.
67. Rosseel Y. lavaan: an R package for structural equation modeling. J Stat Softw. 2012;48:1–36.
68. Letkiewicz AM, Wakschlag LS, Briggs-Gowan MJ, Cochran AL, Wang L, Norton ES, et al. Preadolescent externalizing and internalizing symptoms are differentially related to drift-diffusion model parameters and neural activation during a go/no-go task. J Psychiatr Res. 2024;178:405–13.
69. Zhang T, Yang X, Mu P, Huo X, Zhao X. Stage-specific computational mechanisms of working memory deficits in first-episode and chronic schizophrenia. Schizophr Res. 2025;282:203–13.
70. Gelman A, Rubin DB. Inference from iterative simulation using multiple sequences. Stat Sci. 1992;7:457–72.
71. Lee MD, Wagenmakers E-J Bayesian cognitive modeling: a practical course. Cambridge: Cambridge University Press; 2014.
72. Kruschke JK. Rejecting or accepting parameter values in Bayesian estimation. Adv Methods Pract Psychol Sci. 2018;1:270–80.
73. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in Python. J Mach Learn Res. 2011;12:2825–30.
74. Wu J, Li J, Eickhoff SB, Scheinost D, Genon S. The challenges and prospects of brain-based prediction of behaviour. Nat Hum Behav. 2023;7:1255–64.
75. Aas K, Jullum M, Løland A. Explaining individual predictions when features are dependent: more accurate approximations to Shapley values. Artif Intell. 2021;298:103502.
76. Bora E, Yücel M, Pantelis C. Cognitive impairment in schizophrenia and affective psychoses: implications for DSM-V criteria and beyond. Schizophr Bull. 2010;36:36–42.
77. Owen MJ, O'Donovan MC. Schizophrenia and the neurodevelopmental continuum: evidence from genomics. World Psychiatry. 2017;16:227–35.
78. Zhu Y, Womer FY, Leng H, Chang M, Yin Z, Wei Y, et al. The relationship between cognitive dysfunction and symptom dimensions across schizophrenia, bipolar disorder, and major depressive disorder. Front Psychiatry. 2019;10:253.
79. Pizzagalli DA, Iosifescu D, Hallett LA, Ratner KG, Fava M. Reduced hedonic capacity in major depressive disorder: evidence from a probabilistic reward task. J Psychiatr Res. 2008;43:76–87.
80. Frank MJ. Hold your horses: a dynamic computational role for the subthalamic nucleus in decision making. Neural Netw. 2006;19:1120–36.
81. Seamans JK, Yang CR. The principal features and mechanisms of dopamine modulation in the prefrontal cortex. Prog Neurobiol. 2004;74:1–58.
82. Baek K, Kwon J, Chae J-H, Chung YA, Kralik JD, Min J-A, et al. Heightened aversion to risk and loss in depressed patients with a suicide attempt history. Sci Rep. 2017;7:11228.
83. Bennabi D, Vandel P, Papaxanthis C, Pozzo T, Haffen E. Psychomotor retardation in depression: a systematic review of diagnostic, pathophysiologic, and therapeutic implications. Biomed Res Int. 2013;2013:158746.
84. Buyukdura JS, McClintock SM, Croarkin PE. Psychomotor retardation in depression: biological underpinnings, measurement, and treatment. Prog Neuropsychopharmacol Biol Psychiatry. 2011;35:395–409.
85. Morrens M, Hulstijn W, Sabbe B. Psychomotor slowing in schizophrenia. Schizophr Bull. 2007;33:1038–53.
86. Walther S, Strik W. Motor symptoms and schizophrenia. Neuropsychobiology. 2012;66:77–92.
87. Dunovan K, Verstynen T. Believer-skeptic meets actor-critic: rethinking the role of basal ganglia pathways during decision-making and reinforcement learning. Front Neurosci. 2016;10:106.
88. Mulder MJ, Wagenmakers E-J, Ratcliff R, Boekel W, Forstmann BU. Bias in the brain: a diffusion model analysis of prior probability and potential payoff. J Neurosci. 2012;32:2335–43.
89. Ratcliff R, Huang-Pollock C, McKoon G. Modeling individual differences in the Go/No-go task with a diffusion model. Decision. 2018;5:42–62.
90. Cole MW, Anticevic A, Repovs G, Barch D. Variable global dysconnectivity and individual differences in schizophrenia. Biol Psychiatry. 2011;70:43–50.
91. Zanelli J, Mollon J, Sandin S, Morgan C, Dazzan P, Pilecka I, et al. Cognitive change in schizophrenia and other psychoses in the decade following the first episode. Am J Psychiatry. 2019;176:811–9.
92. Tang C, Huang W, Tan Y, Liu Y, Zheng G, Li B, et al. Comparison of cognitive performance in first-episode drug-naïve schizophrenia, bipolar II disorder, and major depressive disorder patients after treatment. BMC Psychiatry. 2024;24:434.
93. Afshari B, Shiri N, Ghoreishi FS, Valianpour M. Examination and comparison of cognitive and executive functions in clinically stable schizophrenia disorder, bipolar disorder, and major depressive disorder. Depress Res Treat. 2020;2020:2543541.
94. Addington D, Addington J, Schissel B. A depression rating scale for schizophrenics. Schizophr Res. 1990;3:247–51.
95. Addington D, Addington J, Maticka-Tyndale E. Assessing depression in schizophrenia: the Calgary Depression Scale. Br J Psychiatry. 1993;22:39–44.
96. Khandaker GM, Barnett JH, White IR, Jones PB. A quantitative meta-analysis of population-based studies of premorbid intelligence and schizophrenia. Schizophr Res. 2011;132:220–7.
97. Woodberry KA, Giuliano AJ, Seidman LJ. Premorbid IQ in schizophrenia: a meta-analytic review. Am J Psychiatry. 2008;165:579–87.
98. Kuehner C. Why is depression more common among women than among men?. Lancet Psychiatry. 2017;4:146–58.
99. McGrath J, Saha S, Chant D, Welham J. Schizophrenia: a concise overview of incidence, prevalence, and mortality. Epidemiol Rev. 2008;30:67–76.
100. Verbruggen F, Logan GD. Response inhibition in the stop-signal paradigm. Trends Cogn Sci. 2008;12:418–24.

101. Goeleven E, De Raedt R, Baert S, Koster EH. Deficient inhibition of emotional information in depression. J Affect Disord. 2006;93:149–57.
102. Stolicyn A, Steele JD, Seriès P. Conditioned task-set competition: neural mechanisms of emotional interference in depression. Cogn Affect Behav Neurosci. 2017;17:269–89.
103. Joshi YB, Thomas ML, Braff DL, Green MF, Gur RC, Gur RE, et al. Anticholinergic medication burden-associated cognitive impairment in schizophrenia. Am J Psychiatry. 2021;178:838–47.
104. Mancini V, Latreche C, Fanshawe JB, Varvari I, Zauchenberger C-Z, McGinn N, et al. Anticholinergic burden and cognitive function in psychosis: a systematic review and meta-analysis. Am J Psychiatry. 2025;182:349–59.
105. Chouinard G, Margolese HC. Manual for the Extrapyramidal Symptom Rating Scale (ESRS). Schizophr Res. 2005;76:247–65.
106. Mulder MJ, van Maanen L, Forstmann BU. Perceptual decision neurosciences – a model-based review. Neuroscience. 2014;277:872–84.
107. Frank MJ, Gagne C, Nyhus E, Masters S, Wiecki TV, Cavanagh JF, et al. fMRI and EEG predictors of dynamic decision parameters during human reinforcement learning. J Neurosci. 2015;35:485–94.

AUTHOR CONTRIBUTIONS

Tongyi Zhang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. Sirui Wang: Data curation, Investigation. Haoyang Shi: Data curation, Investigation. Jia Li: Methodology, Validation, Writing – review & editing. Xiaolong Yang: Conceptualization, Resources, Investigation. Jin Jiao: Resources, Investigation. Xiaoning Huo: Resources, Investigation. Xin Zhao: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

FUNDING

This work was supported by the National Natural Science Foundation of China (No. 32260207 [to Xin Zhao]), the Lanzhou City Science and Technology Program Project (No. 2022-3-56 [to Xiaolong Yang] and No. 2022-3-55 [to Xiaoning Huo]), the Gansu Provincial Disease Prevention and Control Research Program (No. GSJKKY2025-19 [to Xiaolong Yang]), and the Gansu Provincial Health Industry Research Project (No. GSWSQN2025-25 [to Xiaolong Yang]). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

Ethical standards

The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Declaration of Helsinki of 1975, as revised in 2013.

ADDITIONAL INFORMATION

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41398-026-04345-3.

Correspondence and requests for materials should be addressed to Xin Zhao.

Reprints and permission information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026